Machine Learning

Lecture 13: Computer Vision

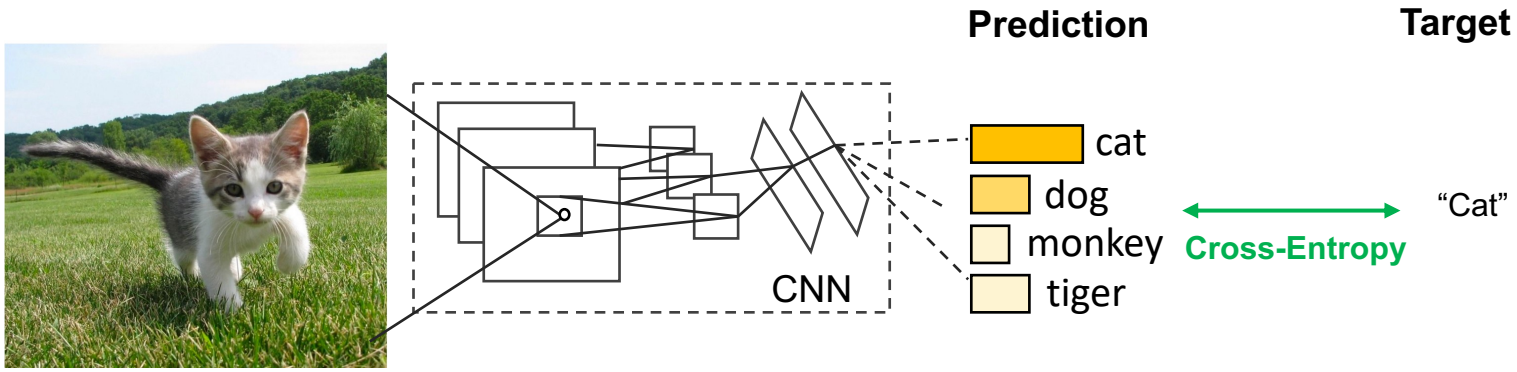
Spring 2025

Instructor: Xiaodong Gu



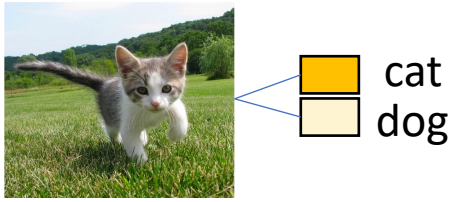
Recall: Image Classification

- Convolution + MLP + Softmax

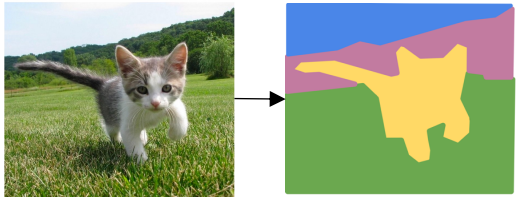


Computer Vision Tasks

Classification



Semantic Segmentation



Object Detection

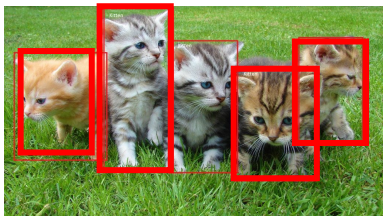


Image Captioning

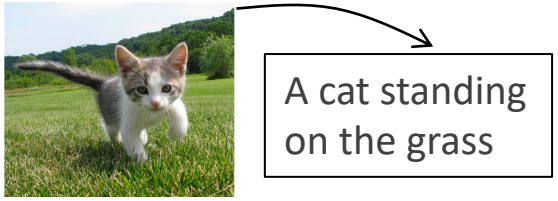
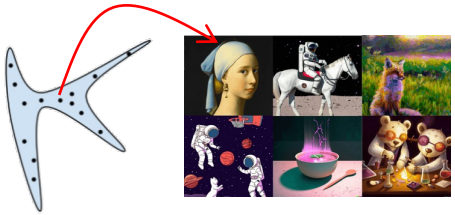


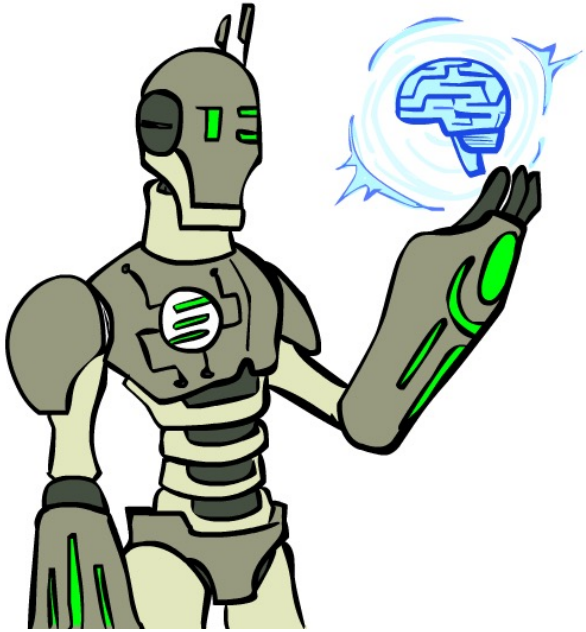
Image Generation



.....

Today

- Image Segmentation
- Object Detection
- Image Captioning

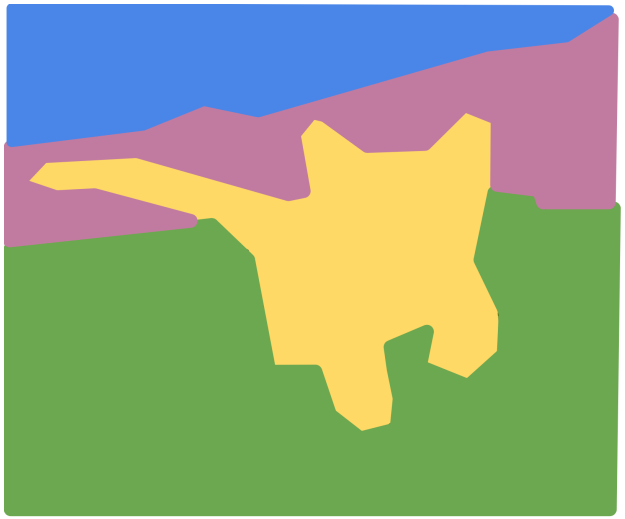


Semantic Segmentation

- Segment an image into disjoint regions that have independent semantics.

Label each pixel in the image with a category





GRASS,
CAT,
TREE,
SKY, ...

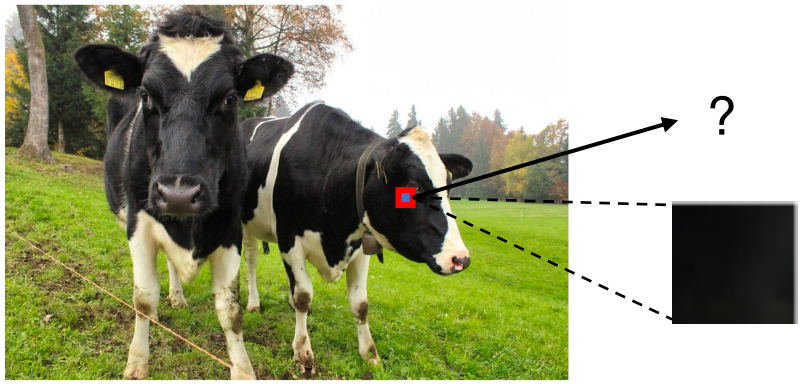
Input: an image

Output: class of each pixel

A Naïve Idea

- **Classifying Pixels?**

Full image

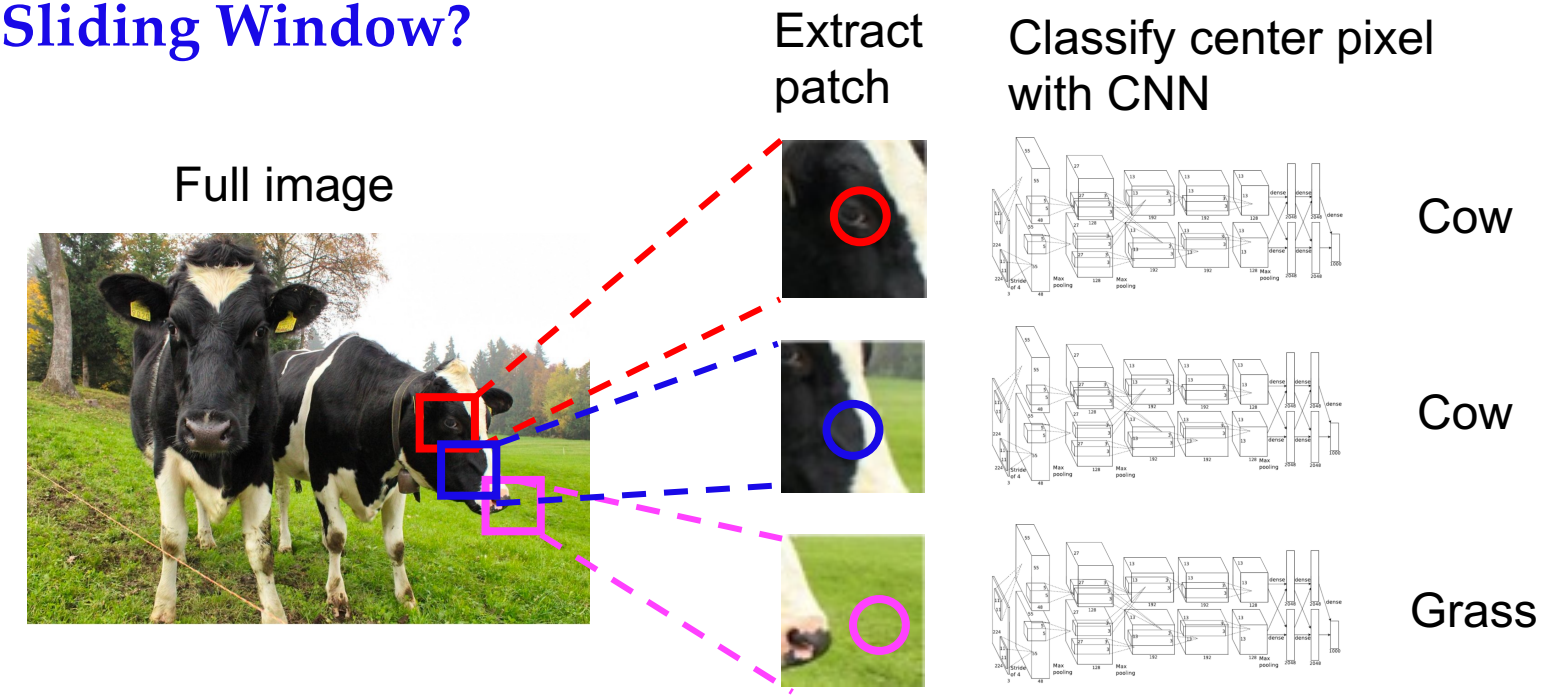


Impossible to classify without context

Q: how do we include context?

The Idea

- **Sliding Window?**

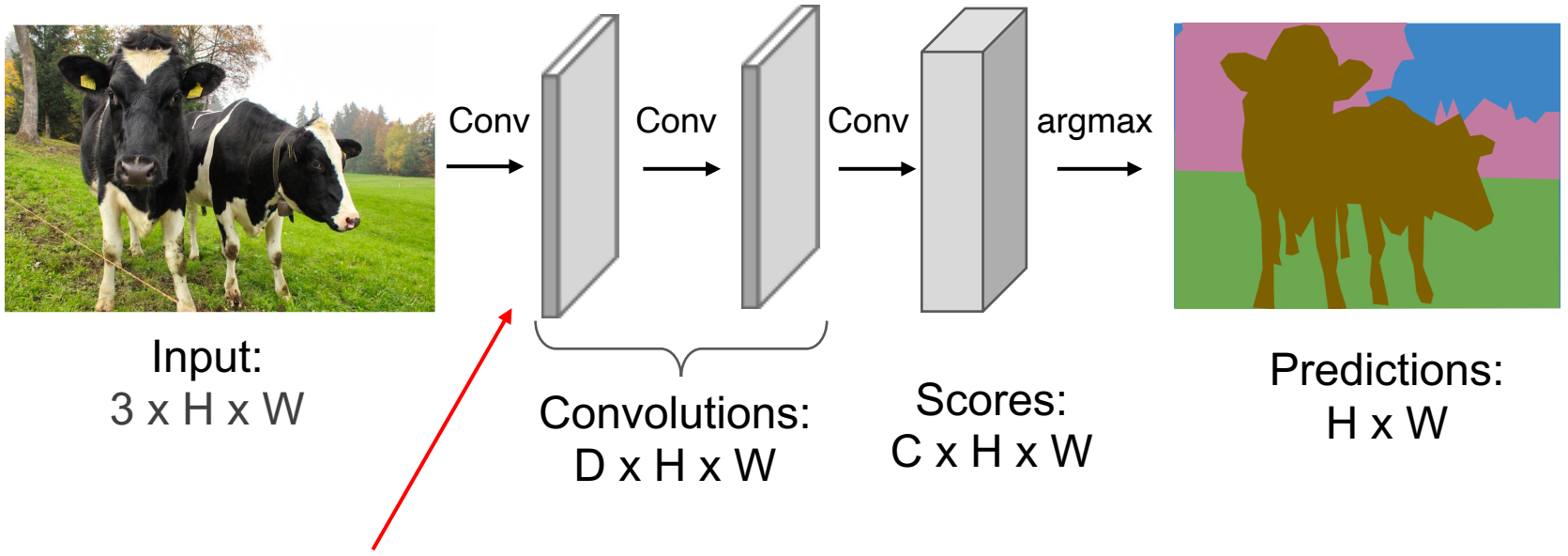


Problem: Very inefficient! Not reusing shared features between overlapping patches

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Fully Convolutional?

- Design a network with only convolutional layers without downsampling operators to make predictions for pixels all at once!

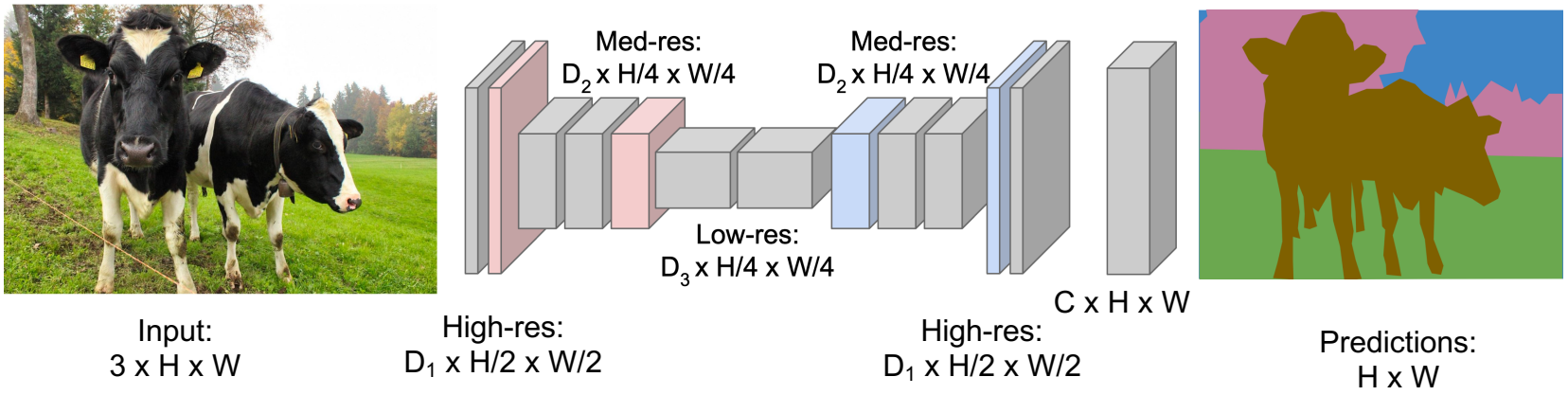


Fully Convolutional?

Instead: Design networks as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Downsampling:
Pooling, strided convolution

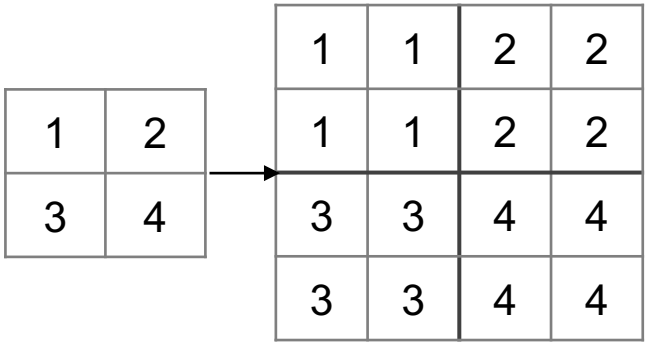
Upsampling:
???



Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

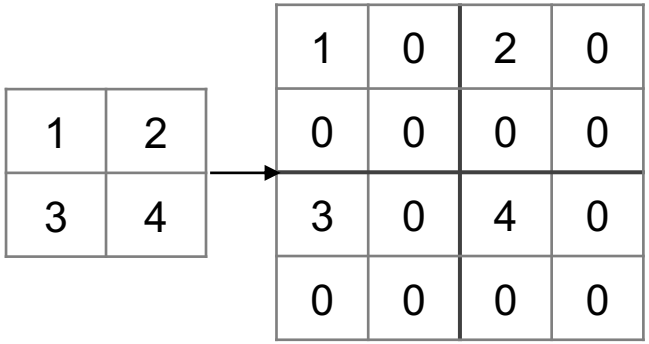
In-Network Upsampling: "Unpooling"

Nearest Neighbor



Input: 2 x 2 Output: 4 x 4

"Bed of Nails"

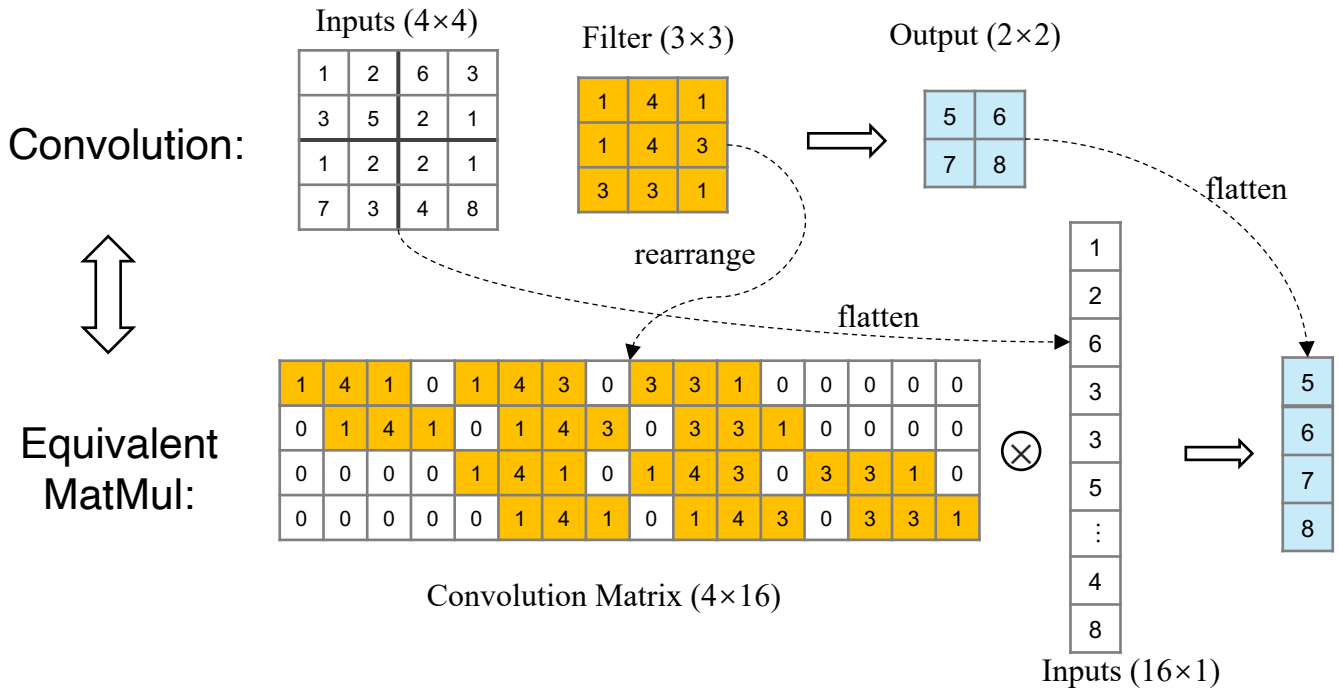


Input: 2 x 2 Output: 4 x 4

Upsampling with Transposed Convolution

- **Convolution Matrix**

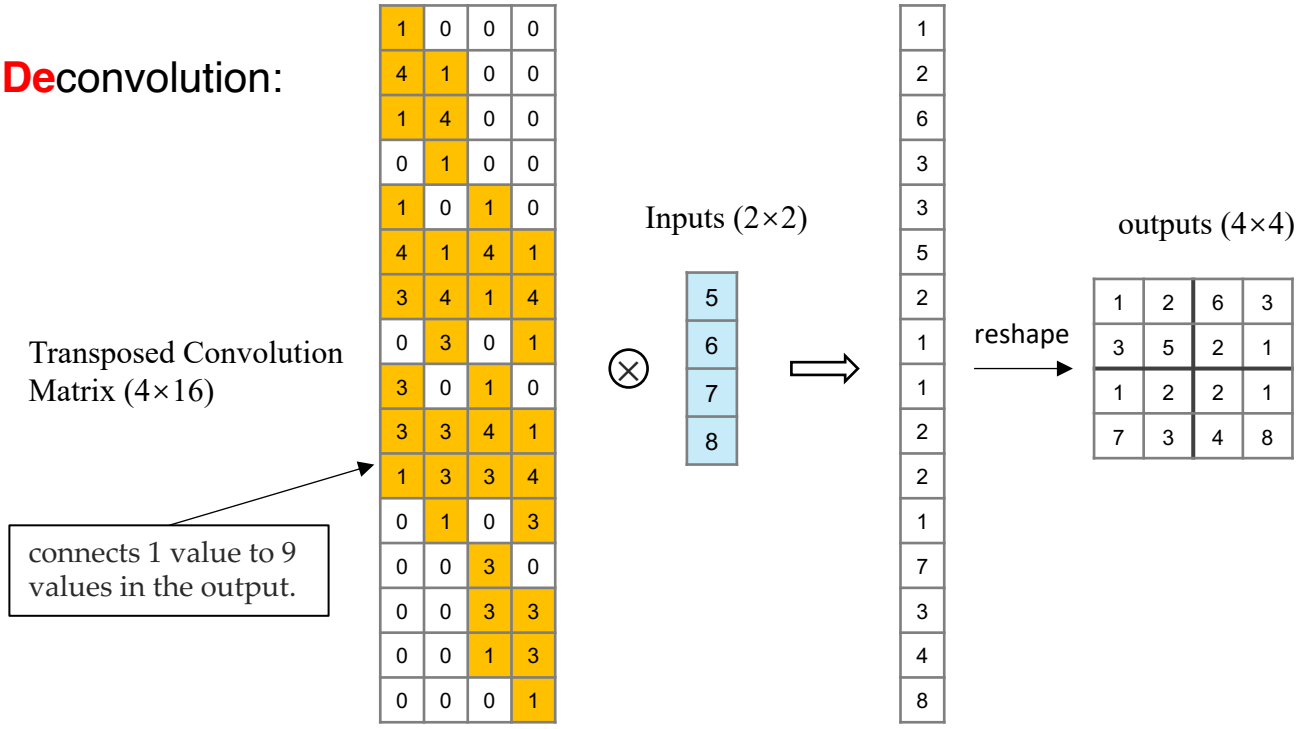
  We can reformulate convolution as **matrix multiplication** by rearranging the kernel matrix into a general matrix.



Upsampling with Transposed Convolution

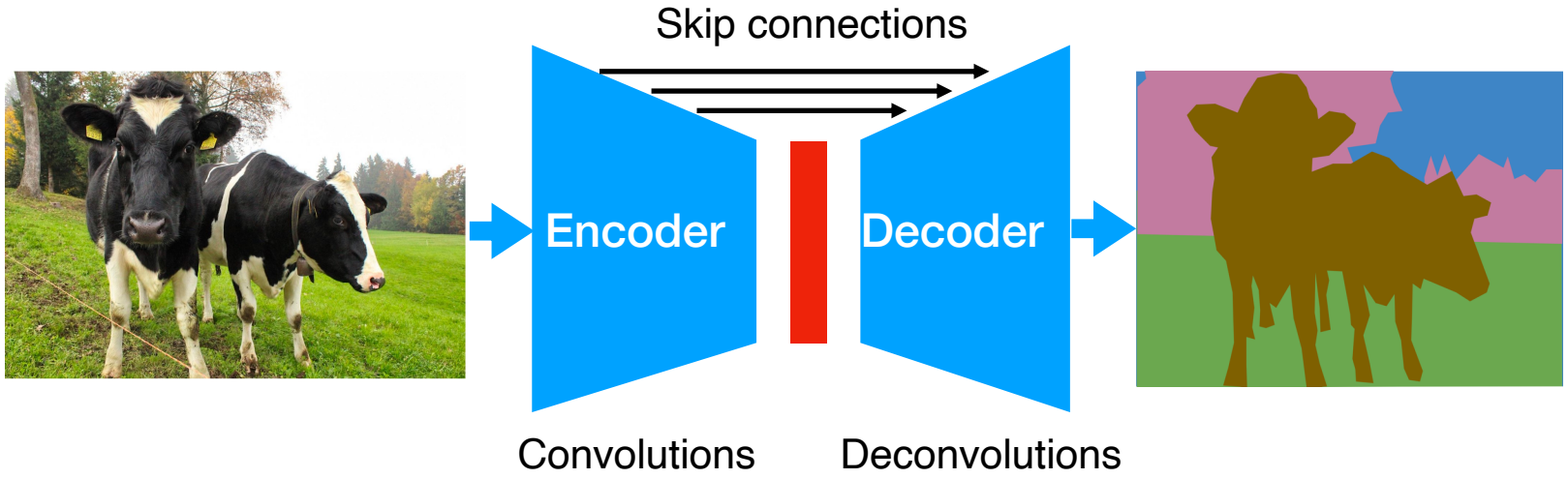
- **Transposed Convolution Matrix**

  Transpose the convolution matrix $\boldsymbol{C}$ (4×16) to $\boldsymbol{C}^{\mathrm{T}}$ (16×4).
  We can matrix-multiply $\boldsymbol{C}^{\mathrm{T}}$ (16×4) with a column vector (4×1) to generate an output matrix (16×1).



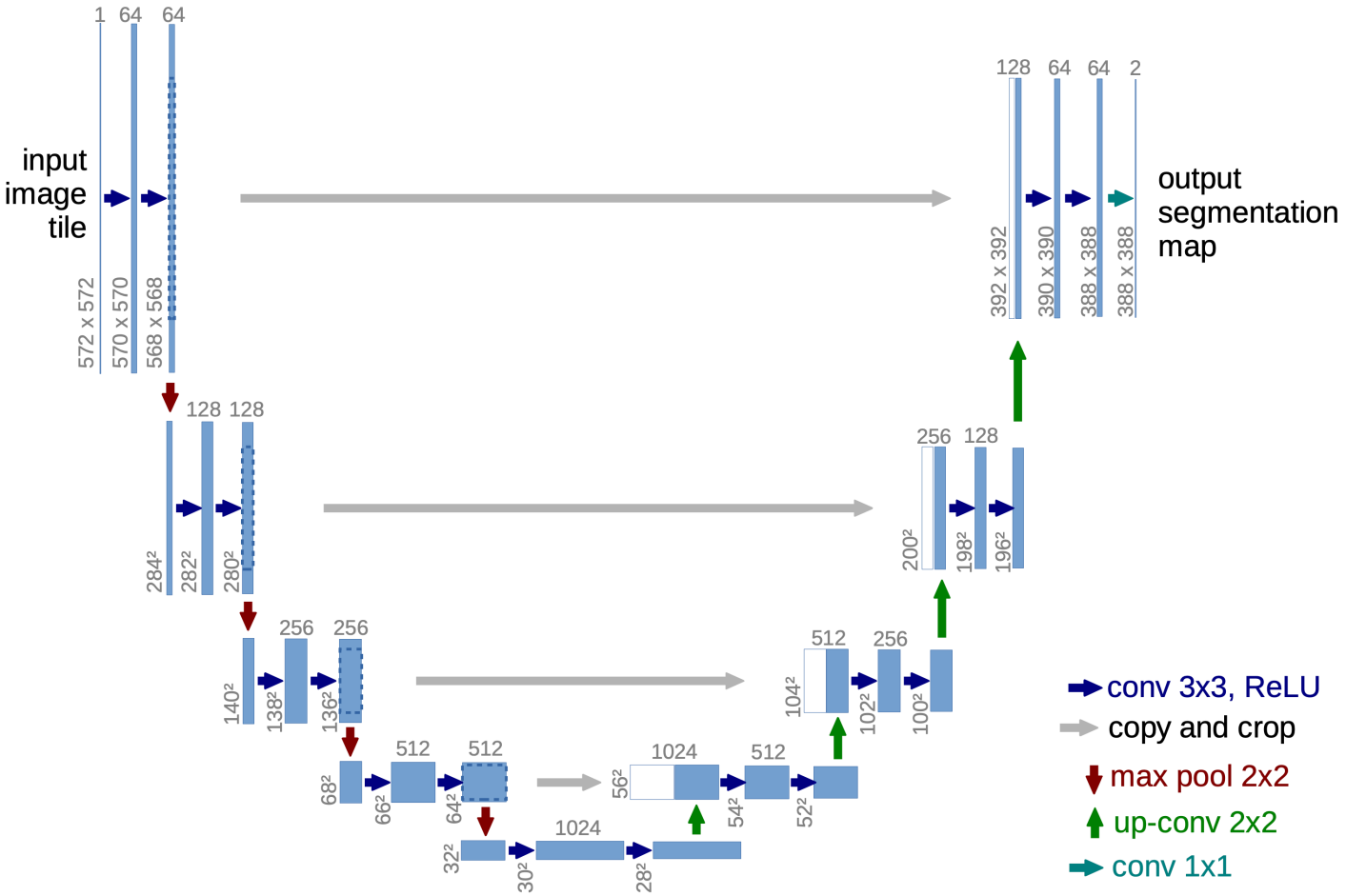
Encoder-Decoder Architectures

- U-Net



Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation

U-Net



Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation

TIME for Coding



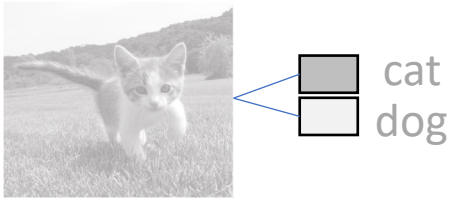
- **Tutorial**: **Image segmentation with U-Net**

  https://www.kaggle.com/code/gokulkarthik/image-segmentation-with-unet-pytorch

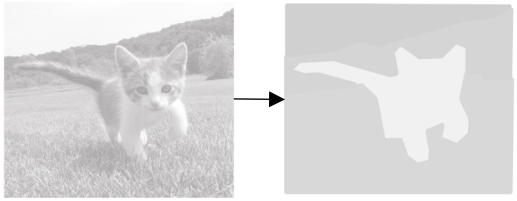
NOTE: How to enable GPU in your Kaggle kernel?
- Verify the phone number of your Kaggle account
- Click "accelerator" in the notebook setting (top right) and then select the GPU (https://www.kaggle.com/general/97939)

Computer Vision Tasks

Classification



Semantic Segmentation



Object Detection

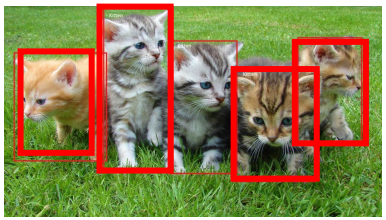
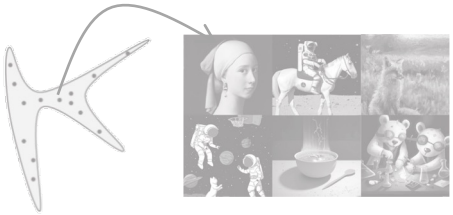


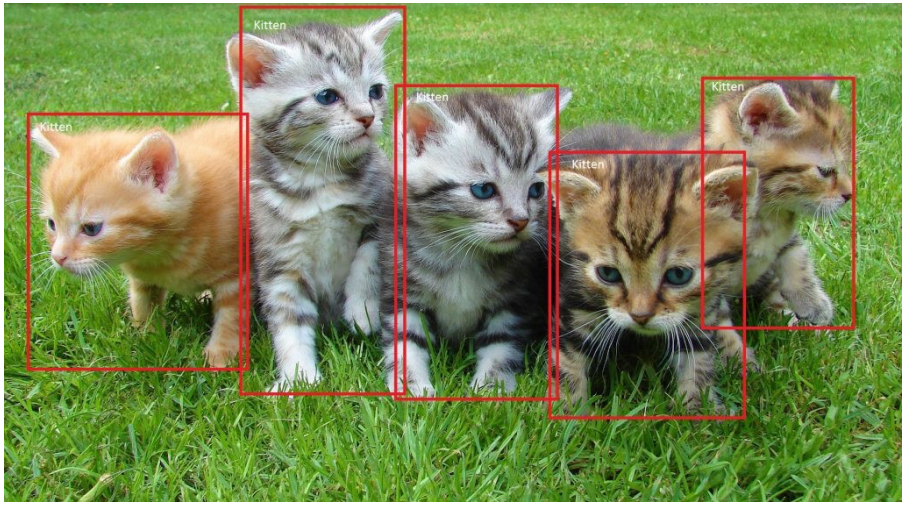
Image Captioning



Image Generation

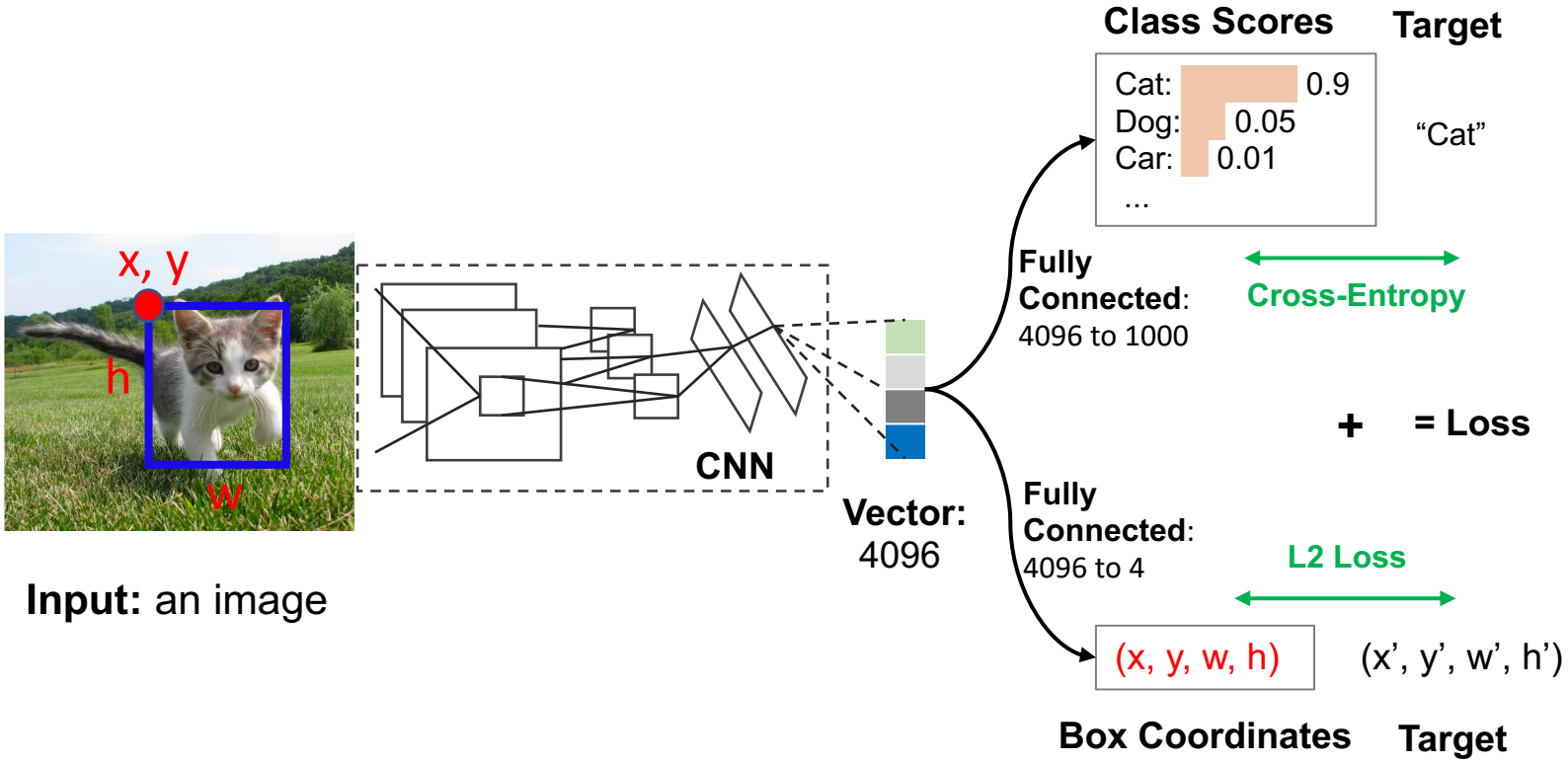


Object Detection

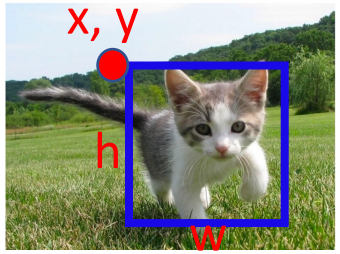


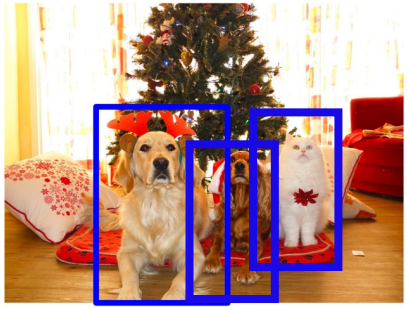
Single Object

- **Object detection = Classification + Localization**

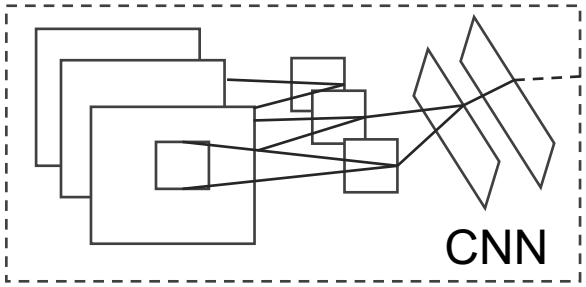


Multiple Objects?

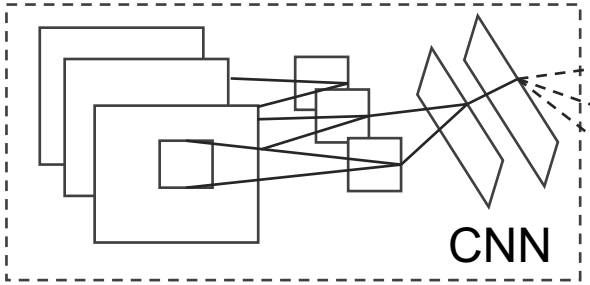




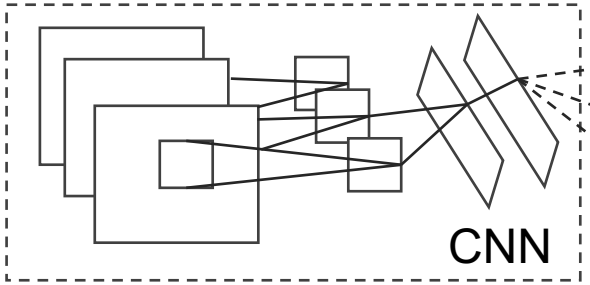




CAT: (x, y, w, h)



DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)



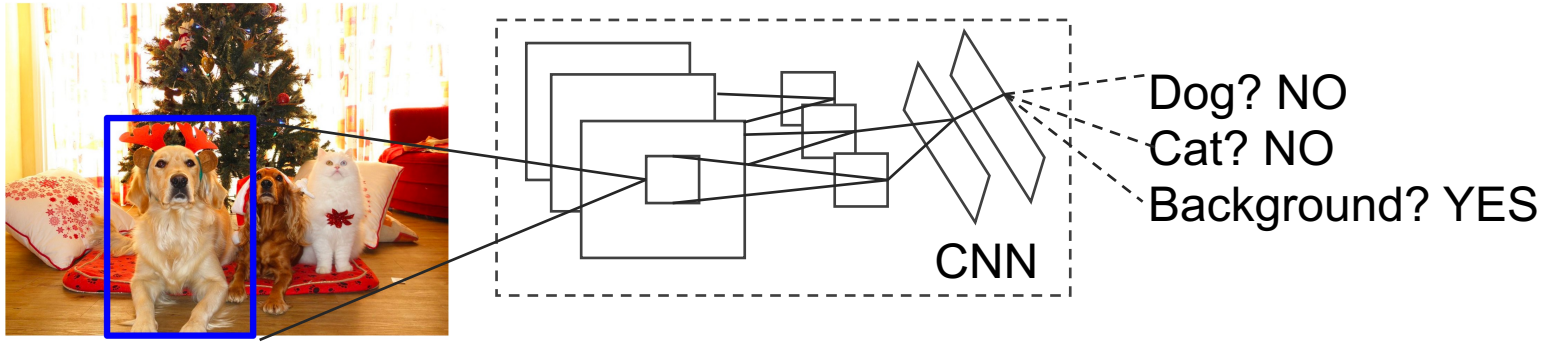
DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
....

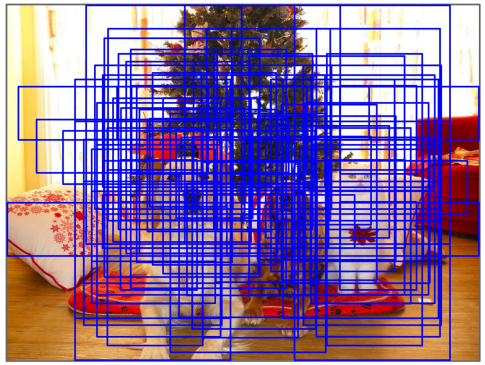
Too many predictions

- **Problem:** each image (CNN) needs a different number of outputs!

Multiple Objects

- Idea: enumerate different crops of the image, apply a CNN to classify **each** crop as object or background.

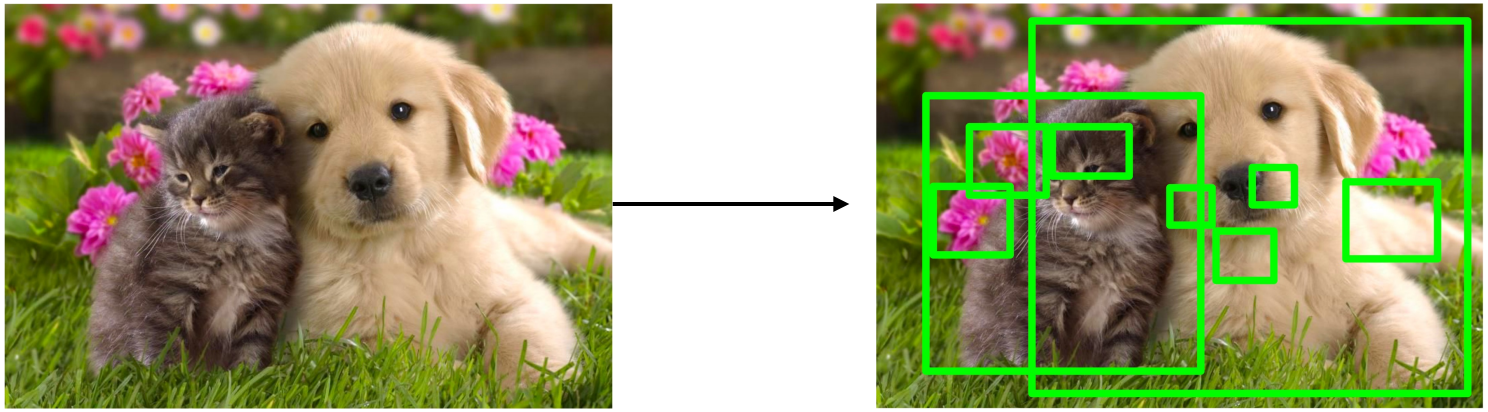




Problem: Need to apply CNN to huge number of locations, scales, and aspect ratios, very computationally expensive!

Region Proposals: Selective Search

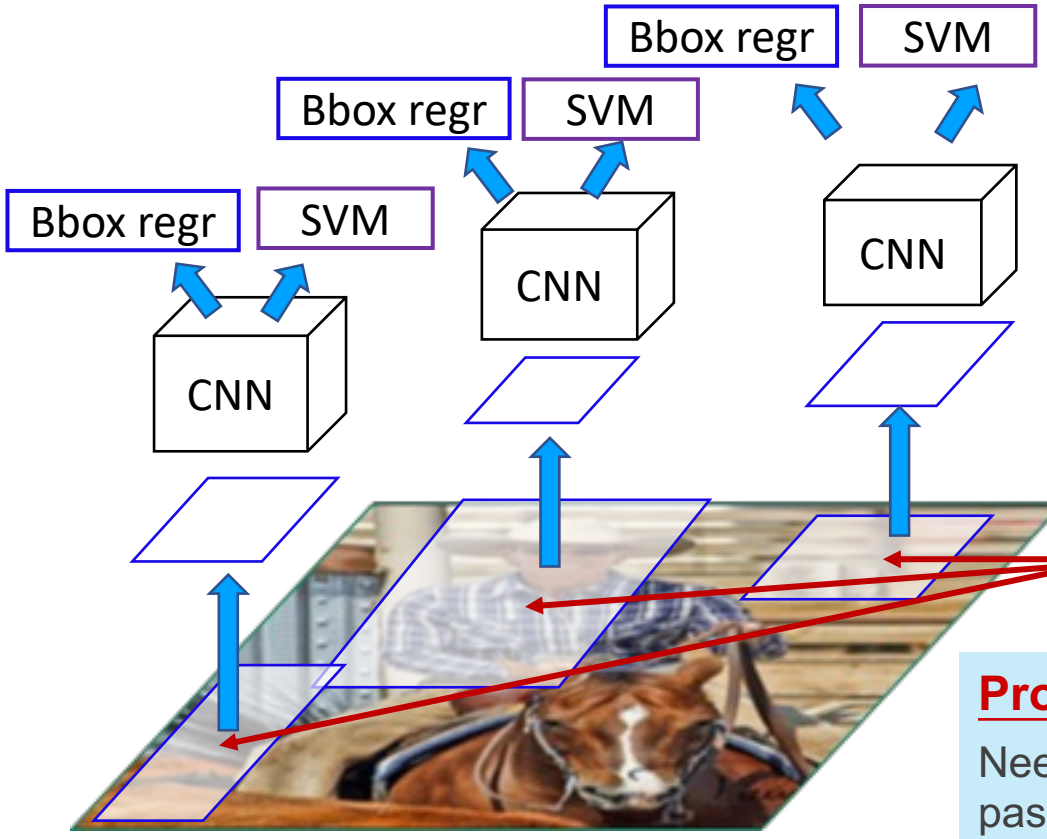
- Find "blobby" image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Alexe et al, "Measuring the objectness of image windows", TPAMI 2012
Uijlings et al, "Selective Search for Object Recognition", IJCV 2013
Cheng et al, "BING: Binarized normed gradients for objectness estimation at 300fps", CVPR 2014
Zitnick and Dollar, "Edge boxes: Locating object proposals from edges", ECCV 2014

R-CNN

⑤ Predict “corrections” to the ROI position: (dx, dy, dw, dh)



④ Classify regions with SVMs

③ Forward each region through ConvNet (ImageNet-pretrained)

② Warped image regions (224x224 pixels)

① **Regions of Interest** (ROI) from a proposal method (~2k)

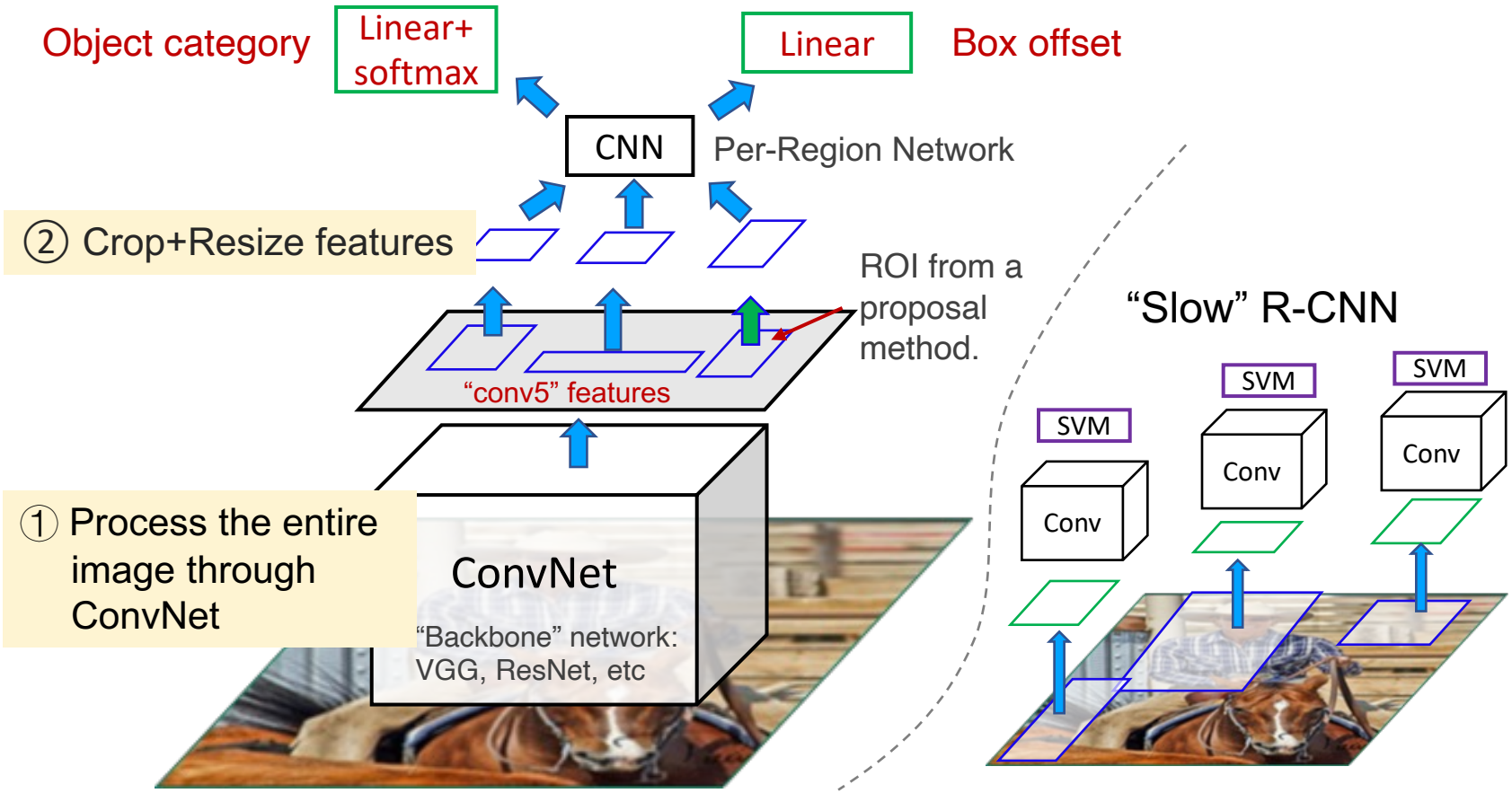
Problem: Very slow!

Need to do ~2k independent forward passes for each image!

Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014.

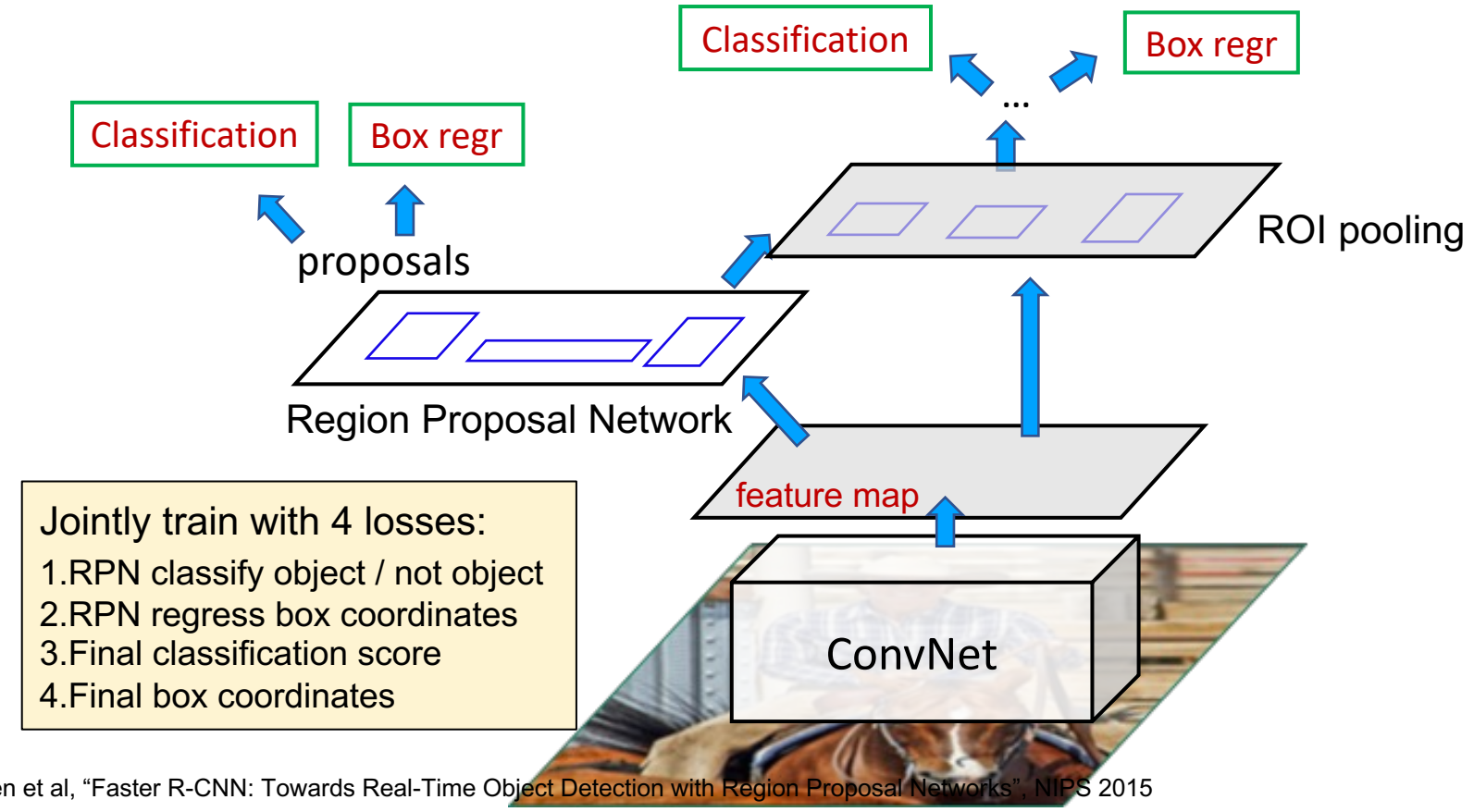
Fast R-CNN Crop the conv feature instead!

- **Idea:** Pass the image through convnet before cropping!



Faster R-CNN Make CNN do proposals !

- Insert a Region Proposal Network (RPN) to predict proposals from features



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015

Region Proposal Network

- Imagine an **anchor box (锚盒)** of fixed size at each point in the feature map



Input Image
(e.g. 3 x 640 x 480)

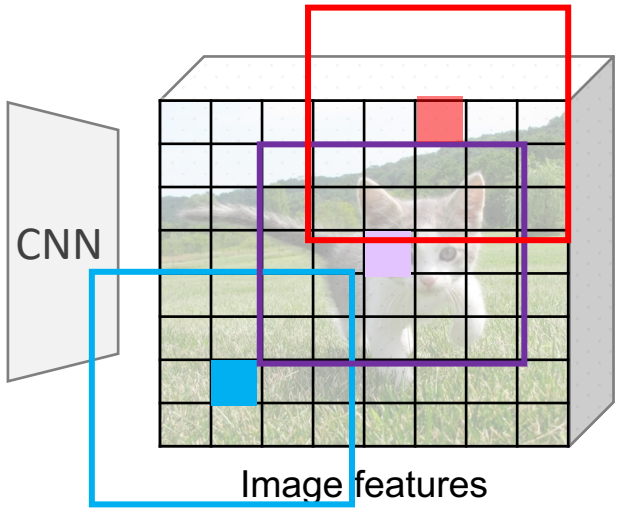
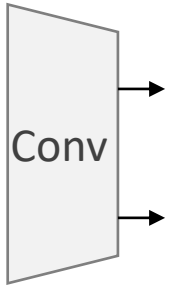


Image features
(e.g. 512 x 20 x 15)

At each point, predict whether the corresponding anchor contains an object (binary classification)



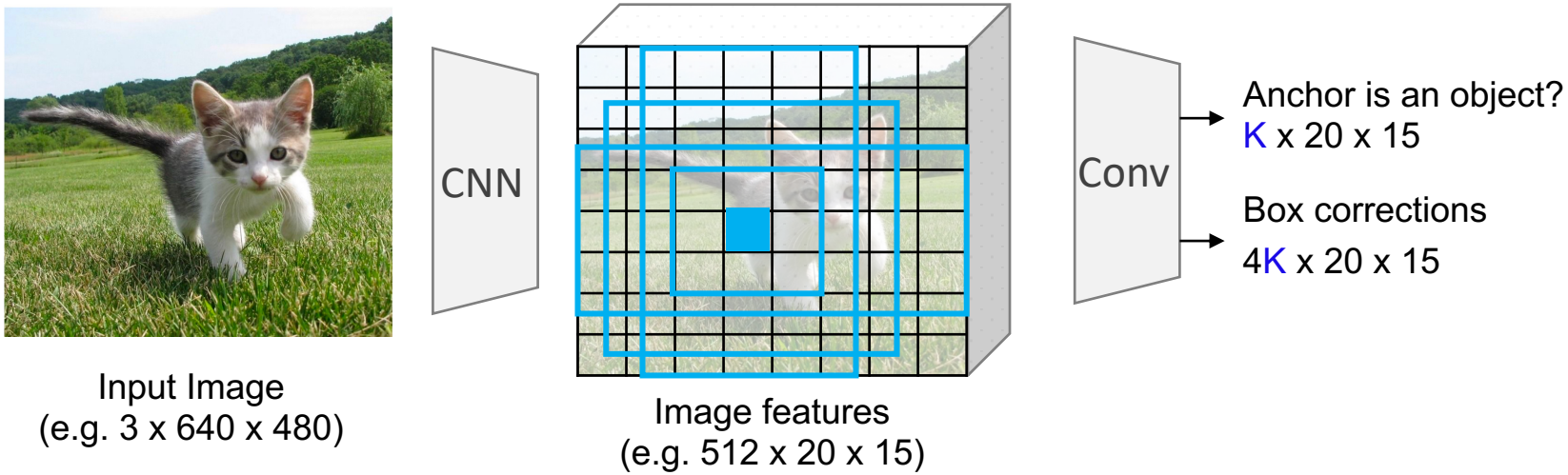
Anchor is an object?
1 x 20 x 15

Box corrections
4 x 20 x 15

For positive boxes, also predict a corrections from the anchor to the ground-truth box (regress 4 numbers per pixel)

Region Proposal Network

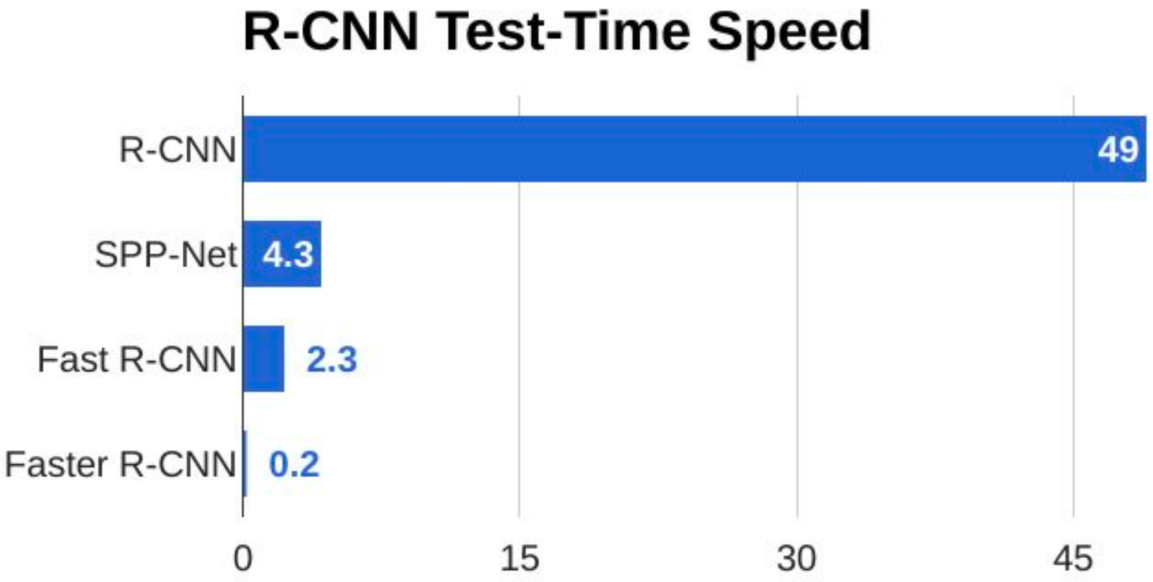
- In practice use K different anchor boxes of different size / scale at each point



- Sort the K*20*15 boxes by their “objectness” score, take top ~300 as our proposals

Faster R-CNN

- How fast?



YOLO – Real Time Image Segmentation

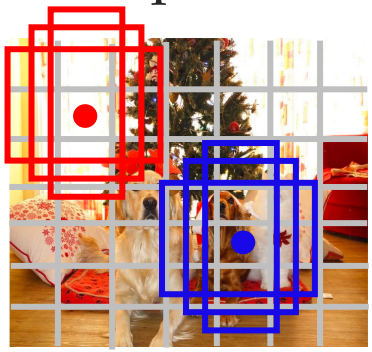
- Formulate object detection as a regression problem



Input image
3 x H x W

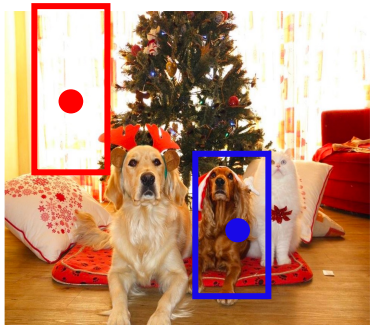


Divide image into SxS (here S=7) grids.



Each grid cell predicts B base boxes with confidence scores: (x, y, h, w, confid).

Regress a final prediction box using the B boxes.



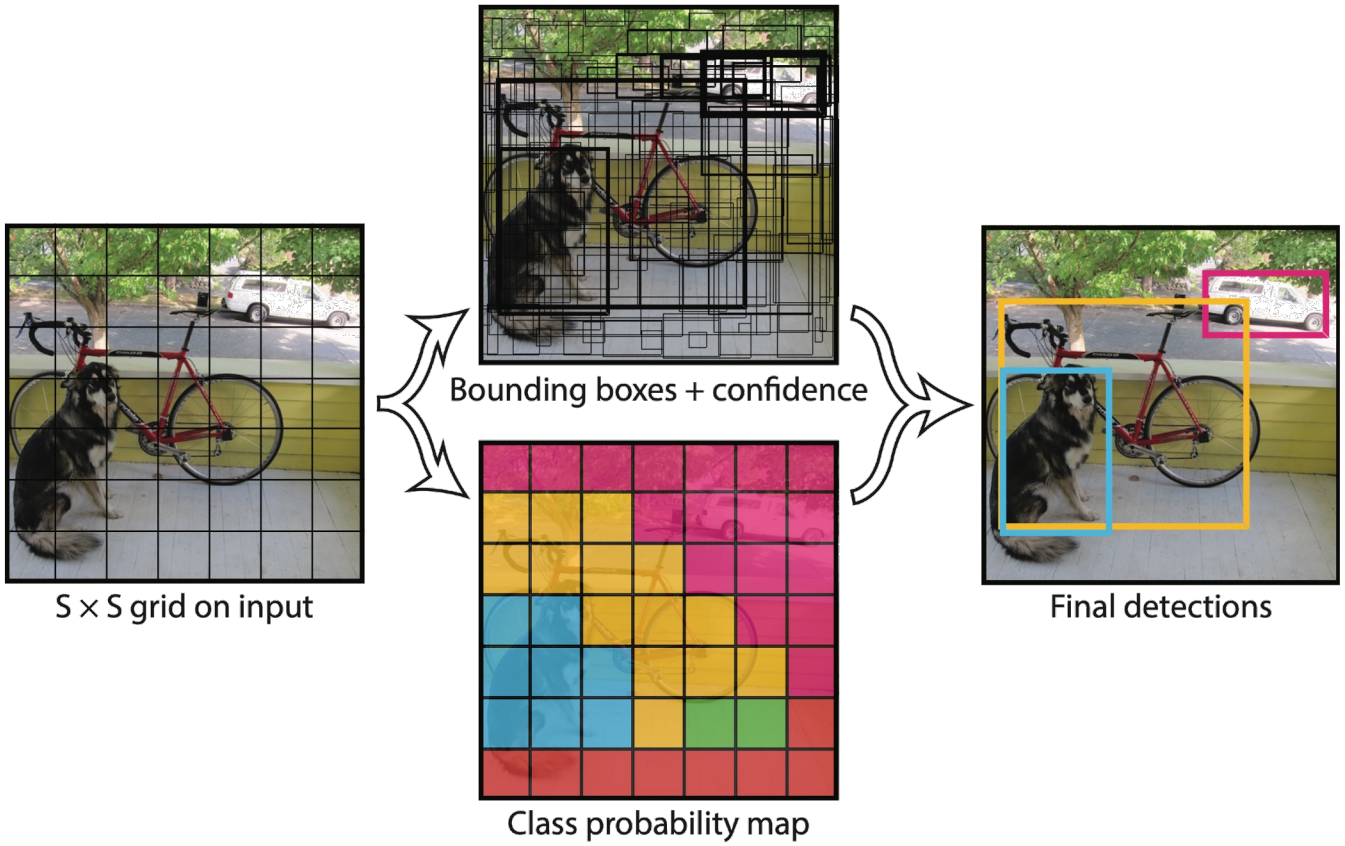


Each cell also predicts probabilities of C classes (including background)

Output:
7 x 7 x (5 * B + C)

Redmon et al. You Only Look Once: Unified, Real-Time Object Detection. CVPR 2016

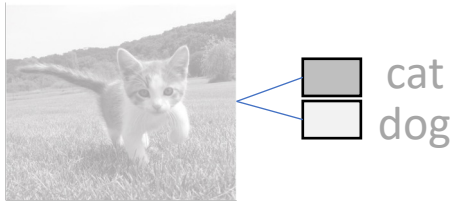
YOLO – Real Time Image Segmentation



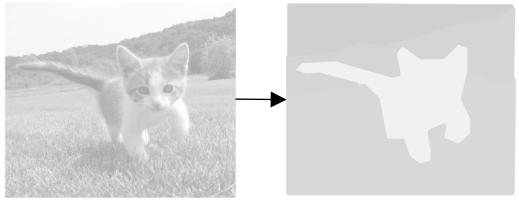
Redmon et al. You Only Look Once: Unified, Real-Time Object Detection. CVPR 2016

Computer Vision Tasks

Classification



Semantic Segmentation



GRASS, CAT,
TREE, SKY, ...

Object Detection

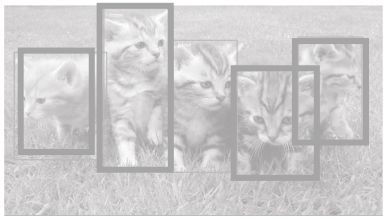


Image Captioning



Image Generation

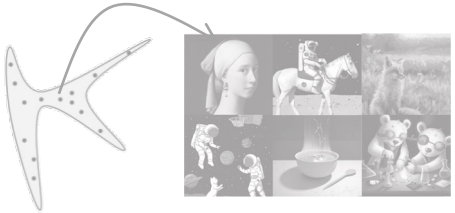


Image Captioning

- Generate a natural language description for an image.



"A person wearing a hat is sitting by the river"

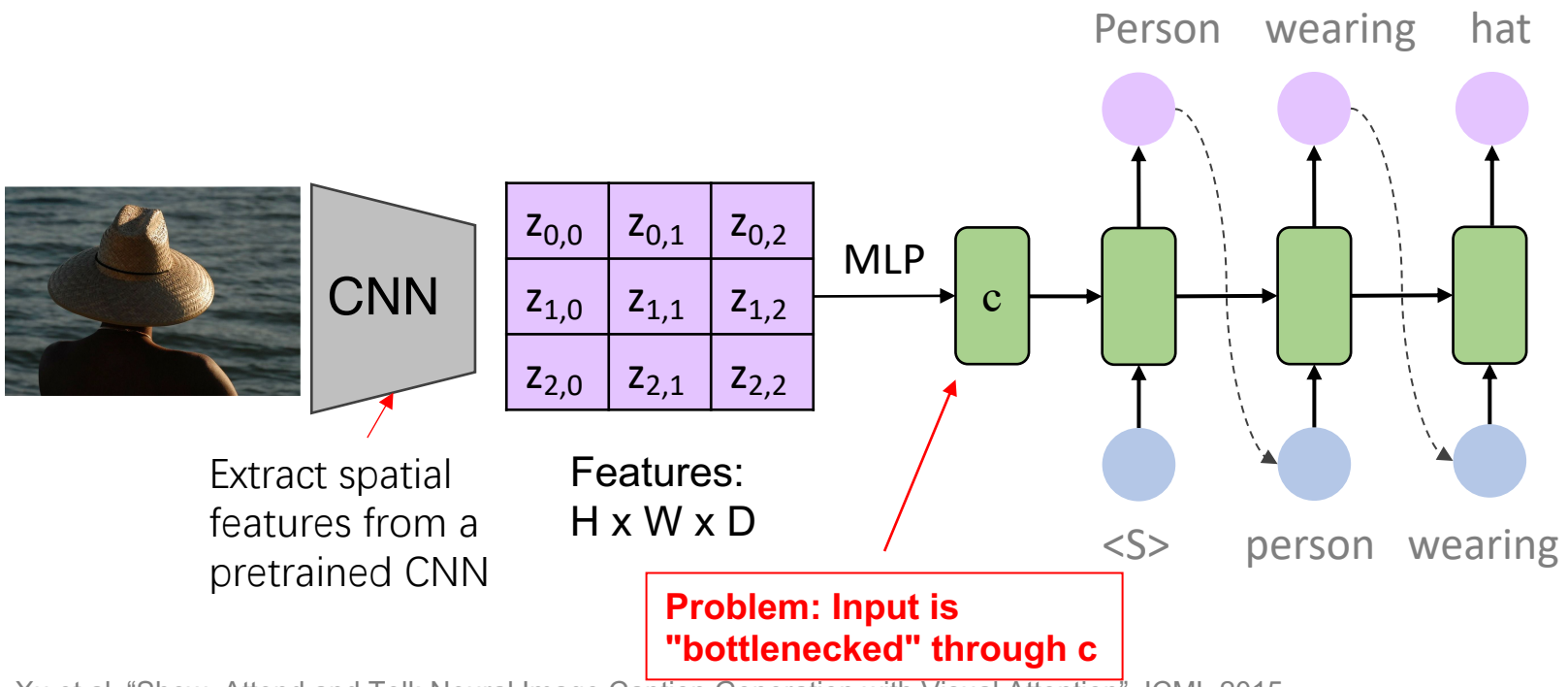
Input: Image **I**

Output: Sequence $\mathbf{y} = y_1, y_2, \ldots, y_T$

Idea: Encoder-Decoder?

Encoder: $c = f_W(\mathbf{z})$
where $\mathbf{z}$ is spatial CNN features
$f_W(.)$ is an MLP

Decoder: $y_t = g_V(y_{t-1}, h_{t-1}, c)$
where context vector c is often $h_0 = c$



Xu et al, "Show, Attend and Tell: Neural Image Caption Generation with Visual Attention", ICML 2015

Encoder-Decoder with Attention?

- Each context vector will attend to different image regions

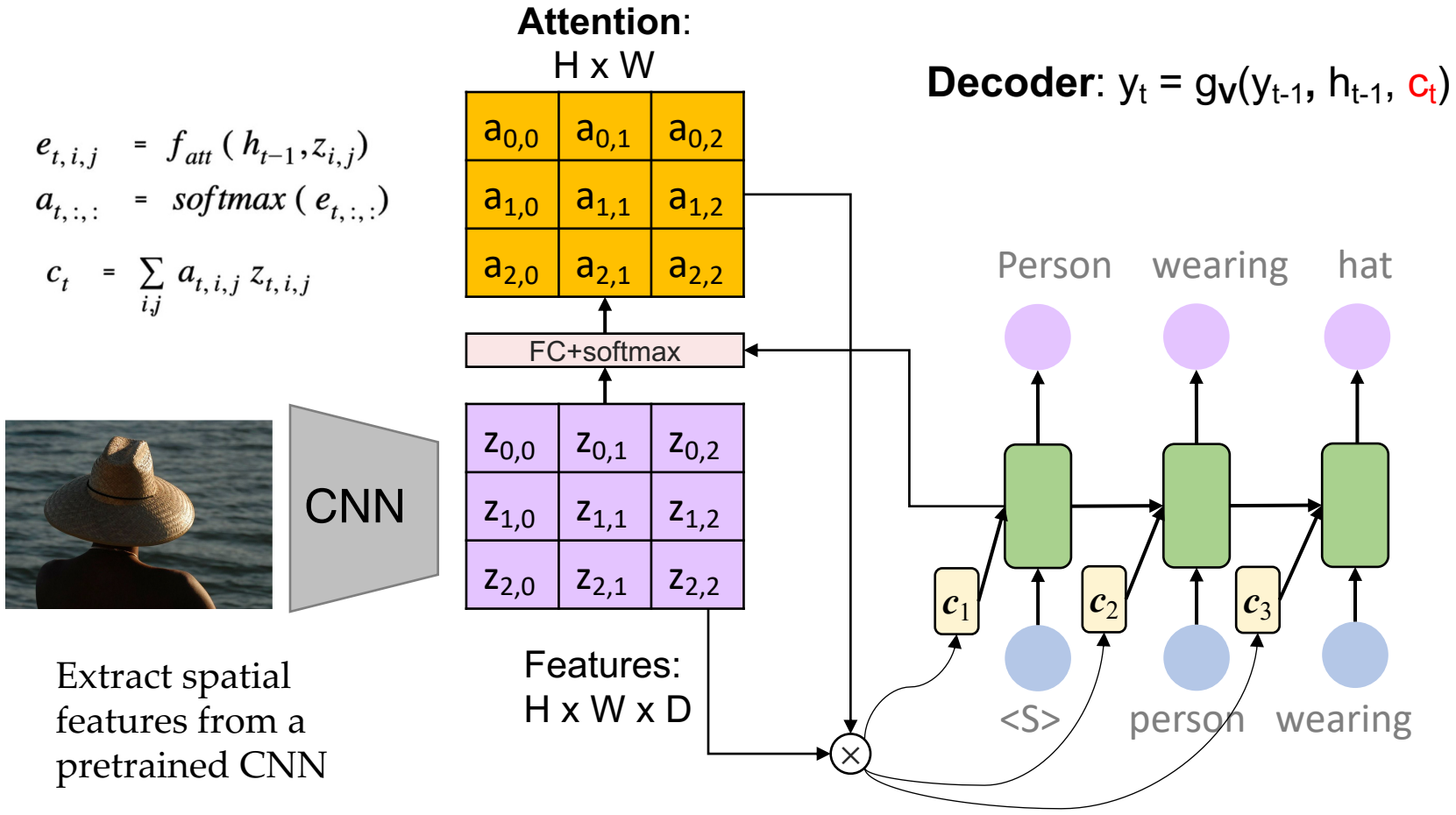


Image Captioning with Attention



A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.

Xu et al, “Show, Attend, and Tell: Neural Image Caption Generation with Visual Attention”, ICML 2015

CNN with Self-Attention

- We can also apply **self-attention** to CNN features.

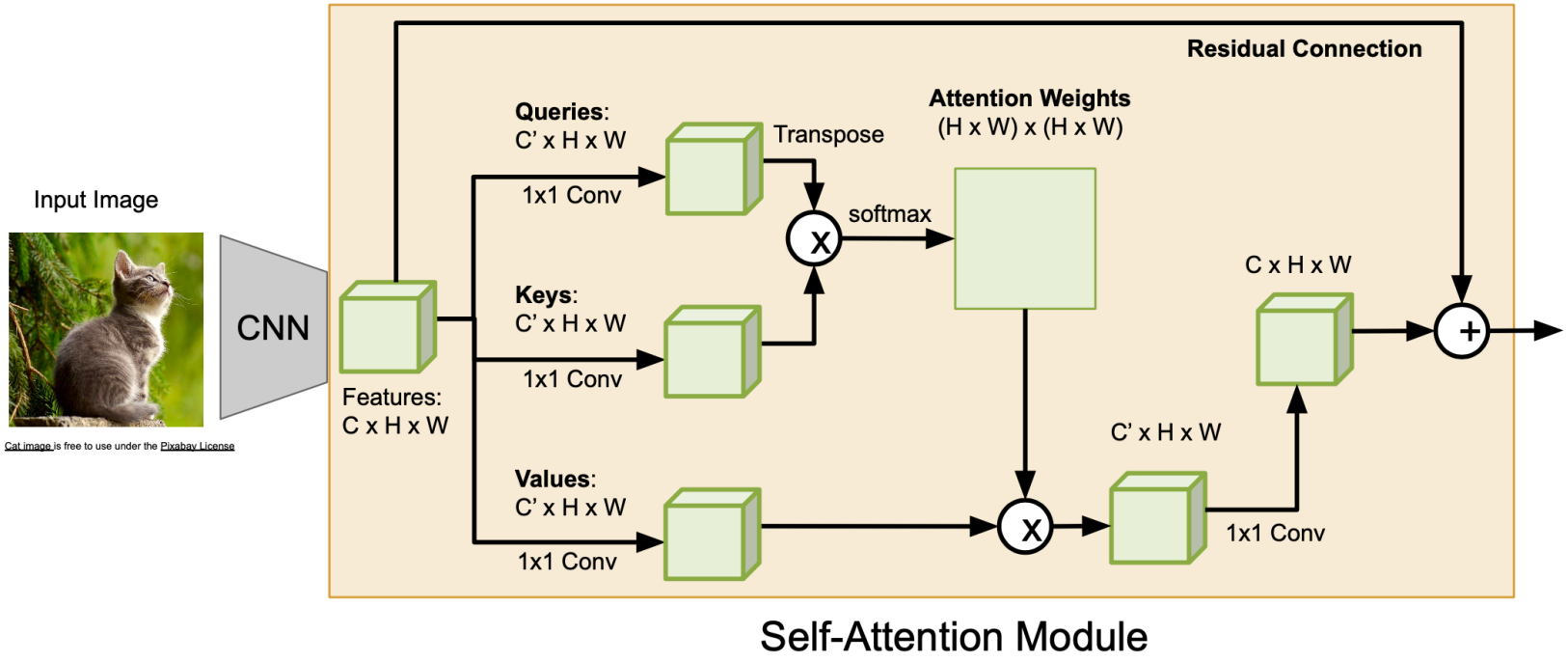
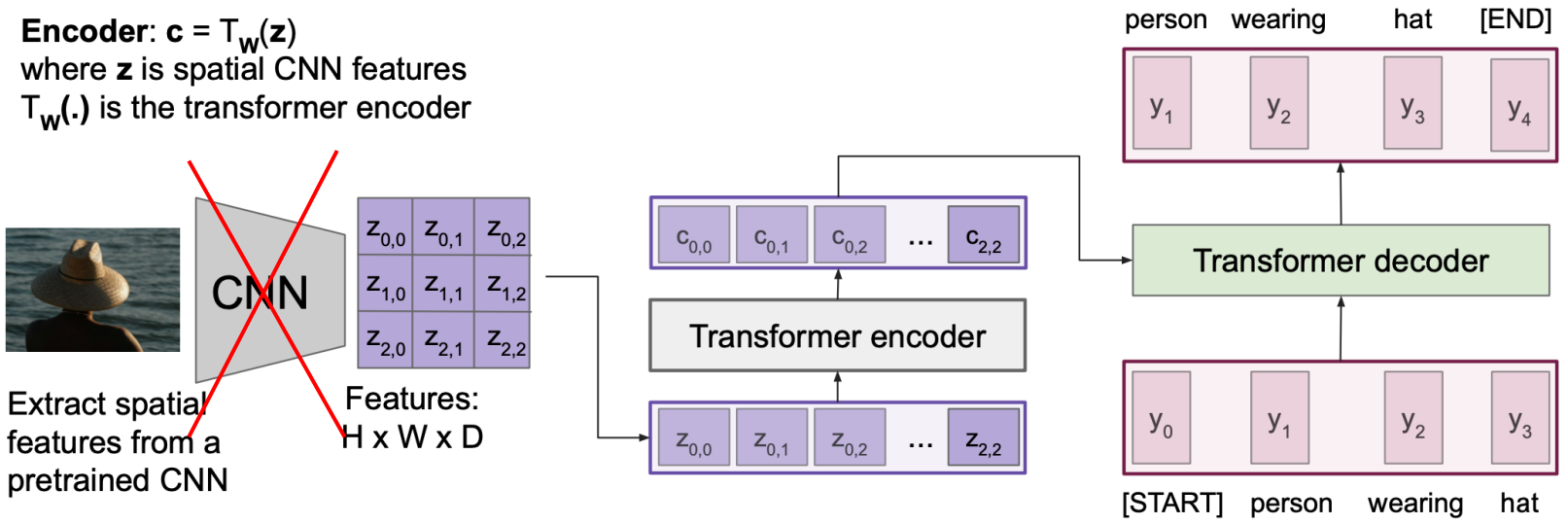
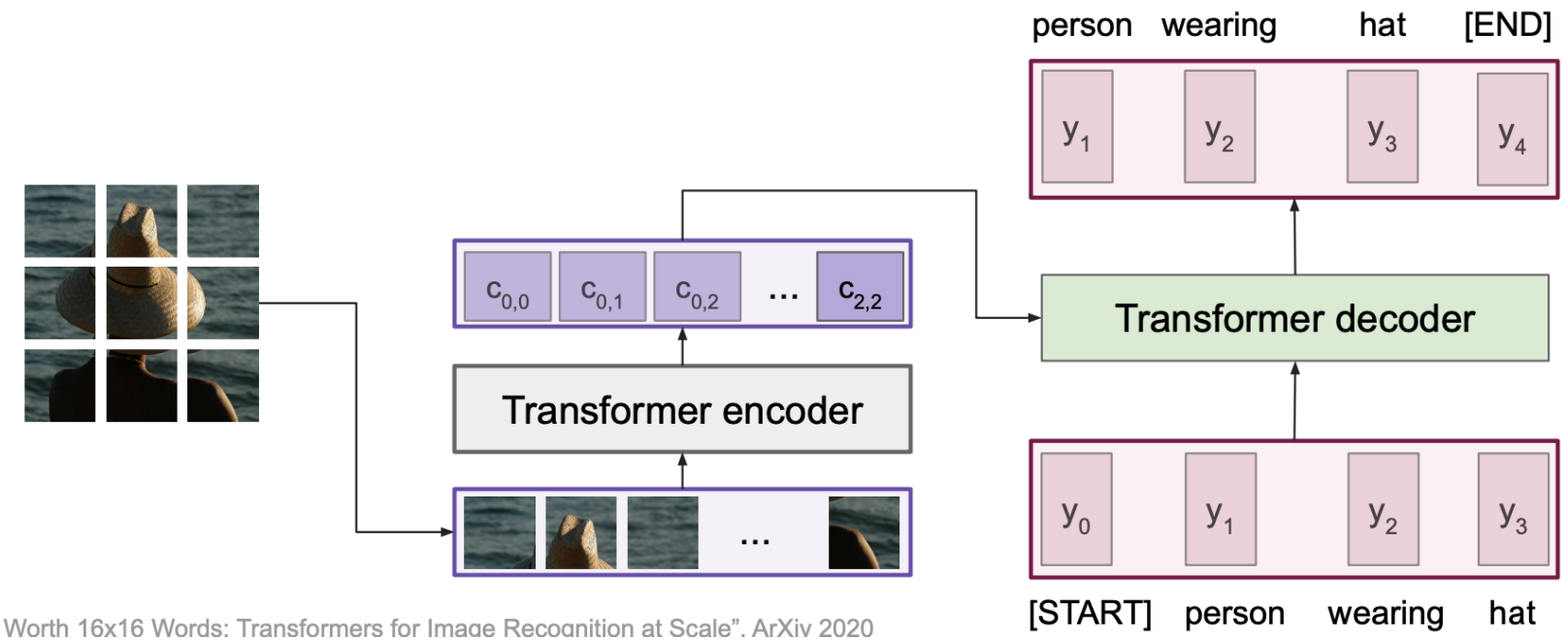


Image Captioning ~~with~~using Transformers



Perhaps we don't need convolutions at all?

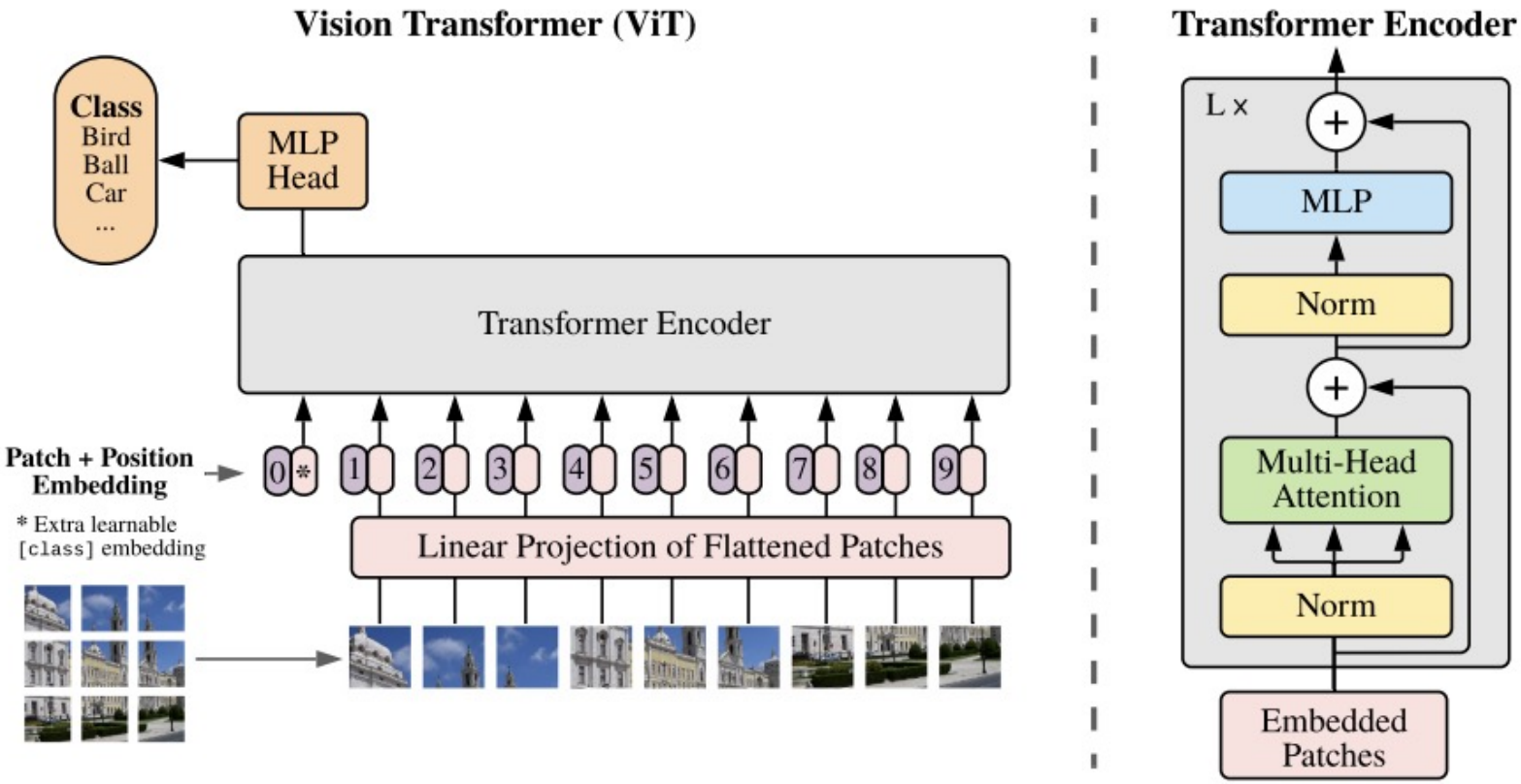
Image Captioning using ONLY Transformers



Worth 16x16 Words: Transformers for Image Recognition at Scale". ArXiv 2020

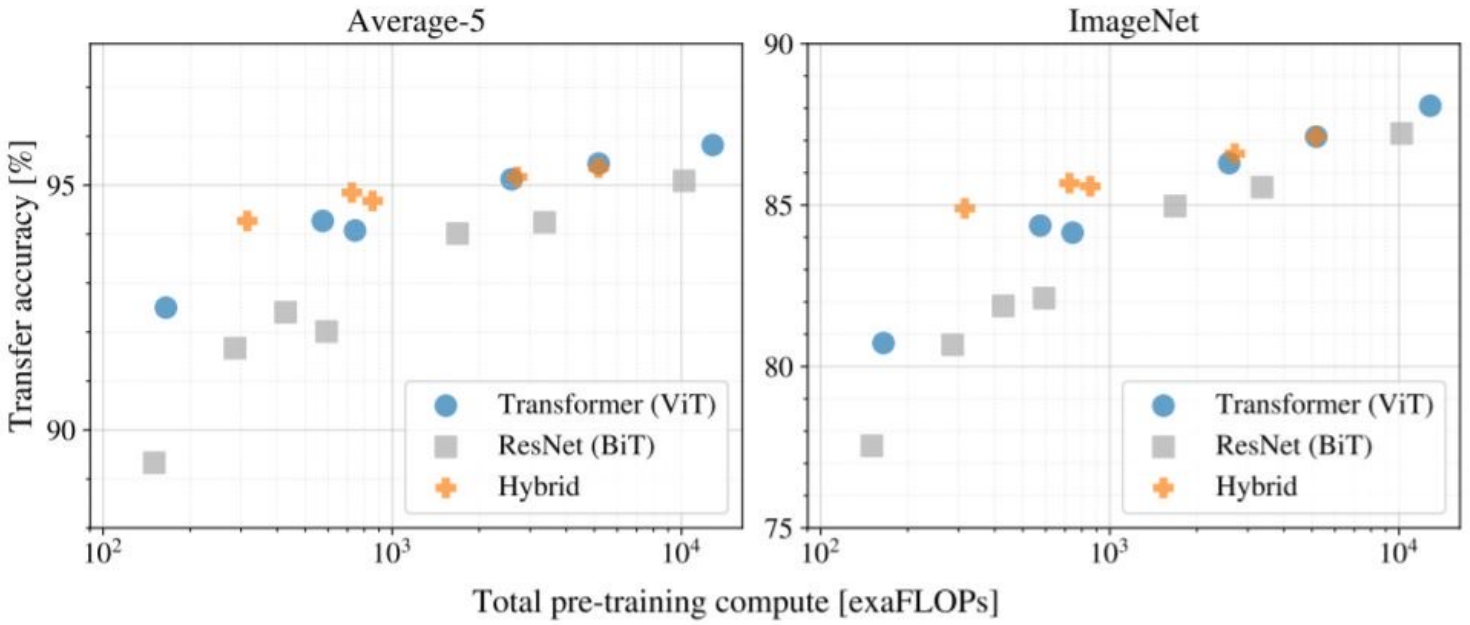
Dosovitskiy et al, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ArXiv 2020

Vision Transformer (ViT)



Dosovitskiy et al, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ArXiv 2020

Vision Transformers vs. ResNets

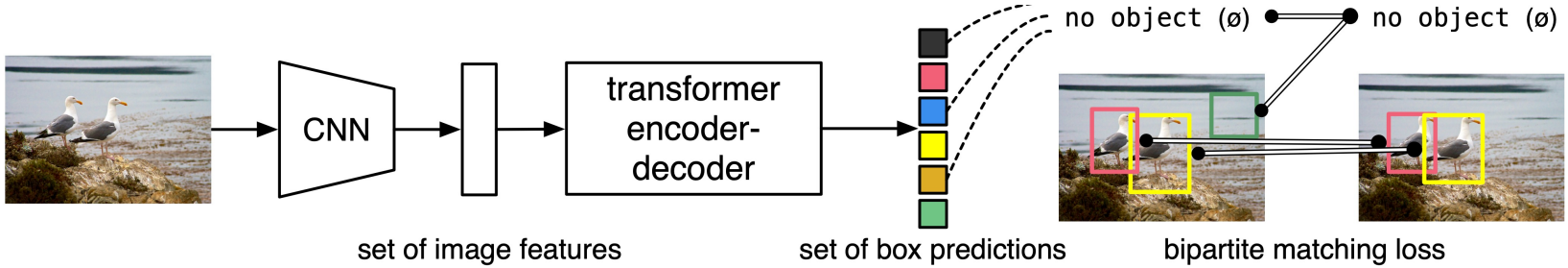


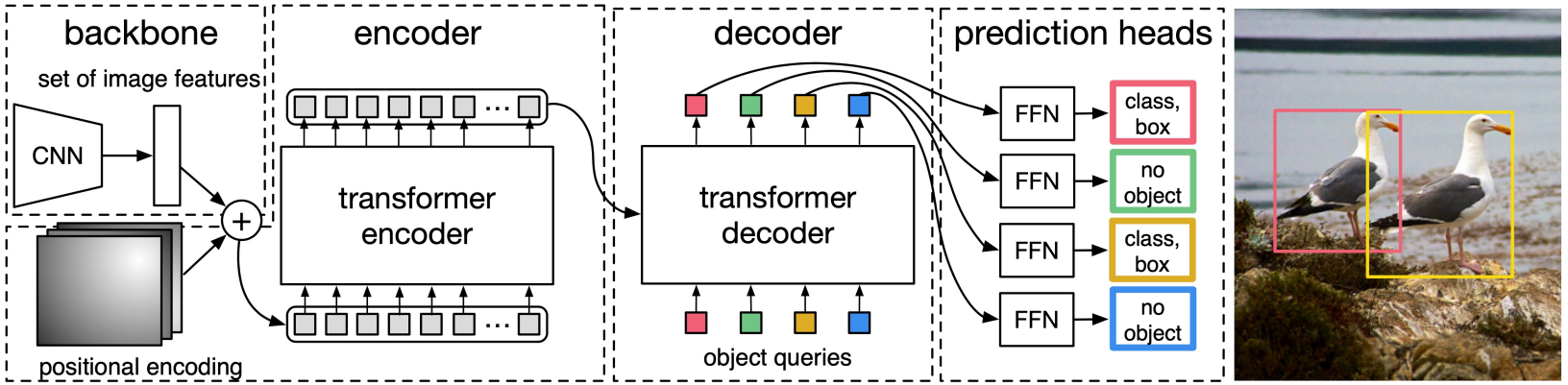
Farewell CNN ?



Vision Transformers for Object Detection

- DE:TR

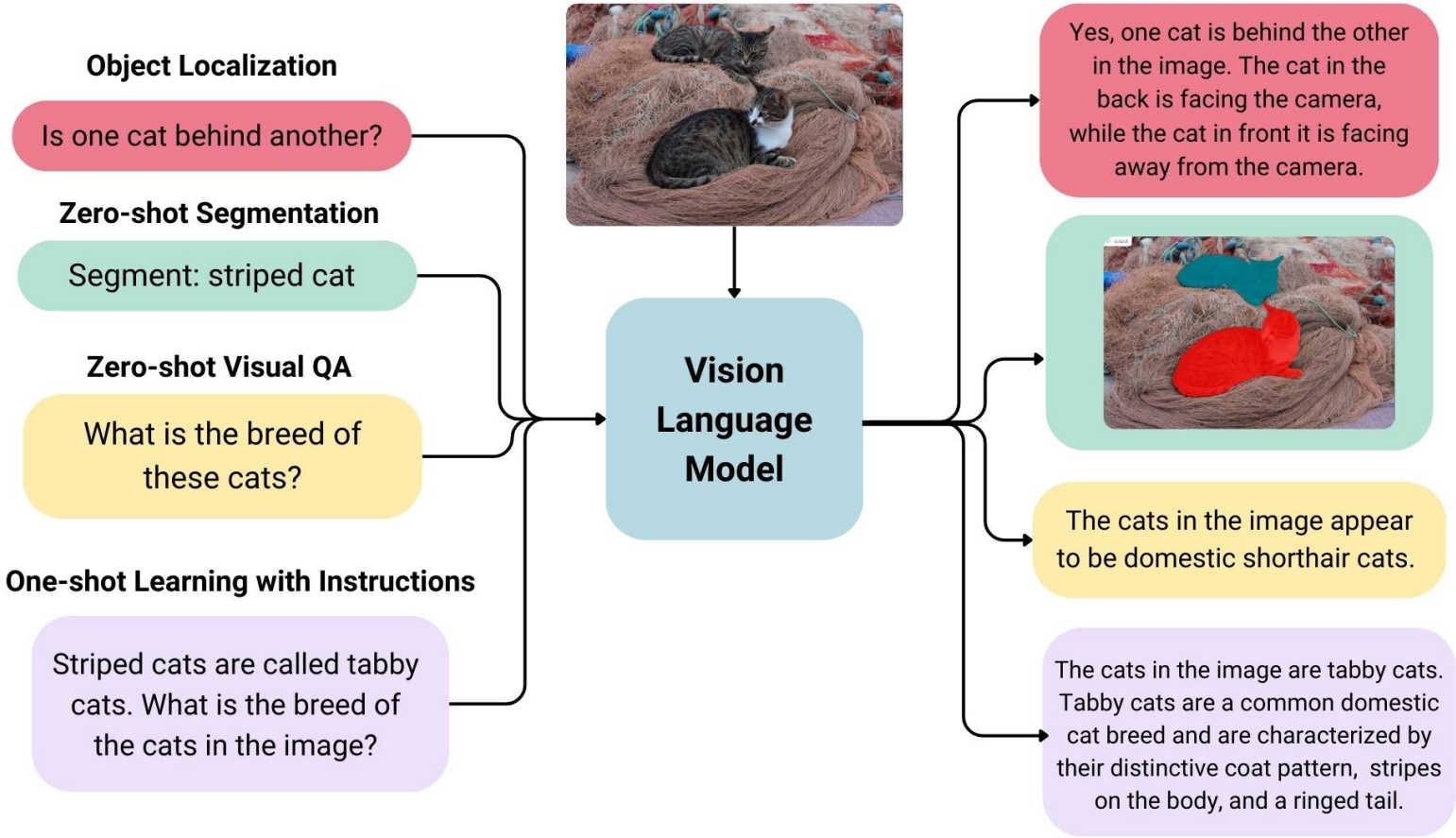




Carion et al, "End-to-End Object Detection with Transformers", ECCV 2020

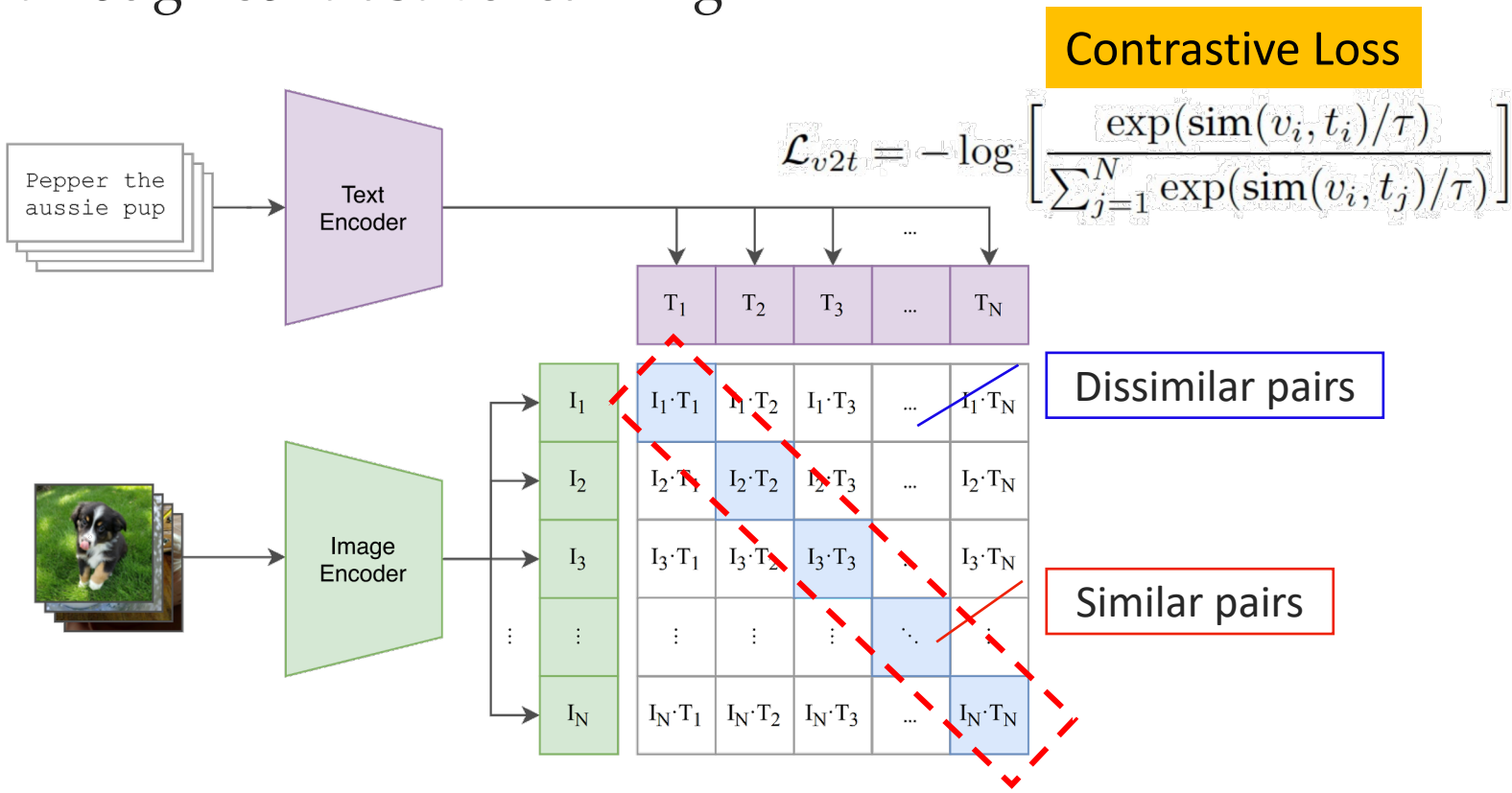
Vision Language Models (VLMs)

- **Language Model for Multimodal (text-image) Data**



CLIP – Contrastive Language-Image Pretraining

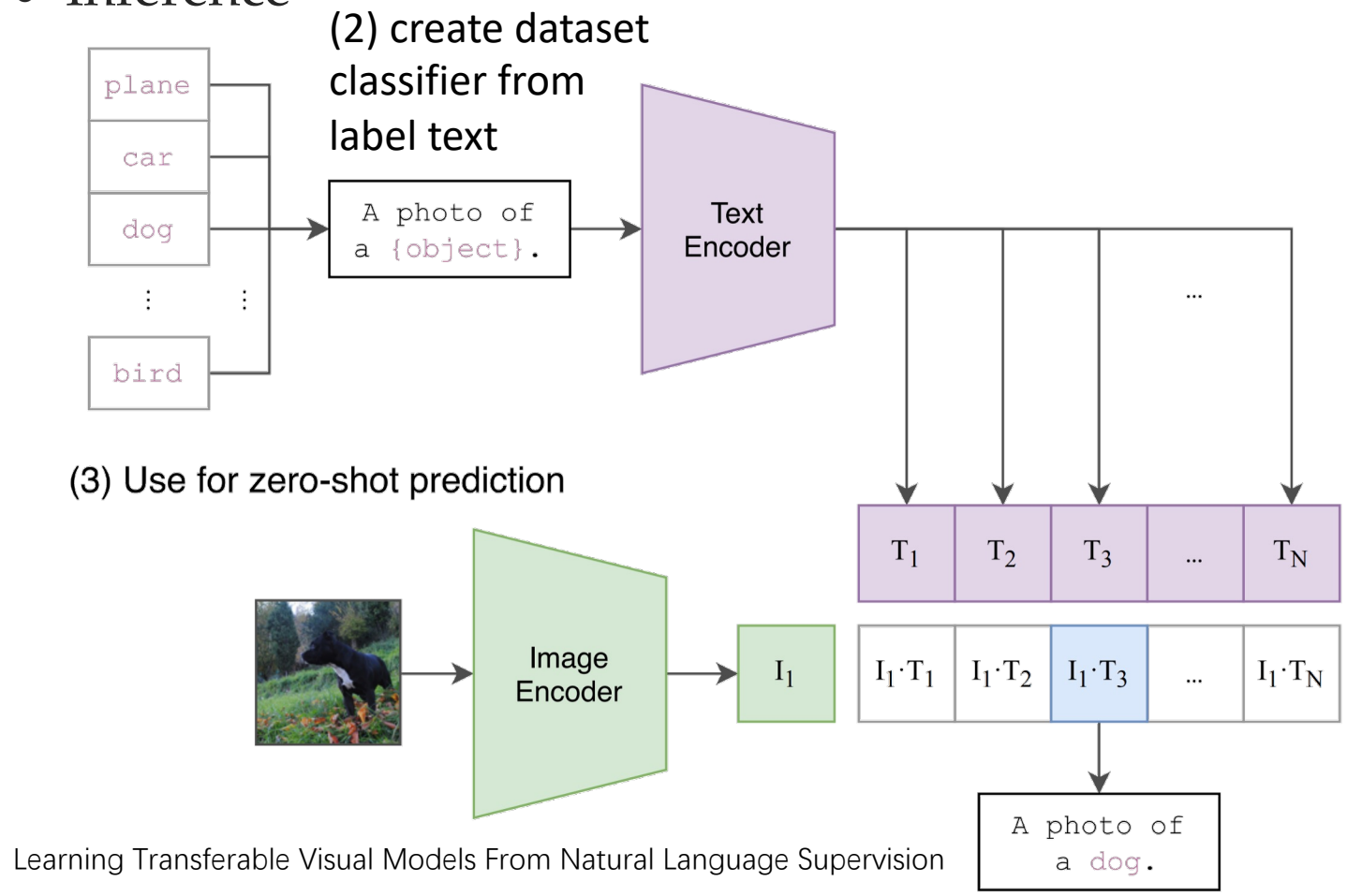
- Encode text and image into the same embedding space through contrastive learning



Learning Transferable Visual Models From Natural Language Supervision

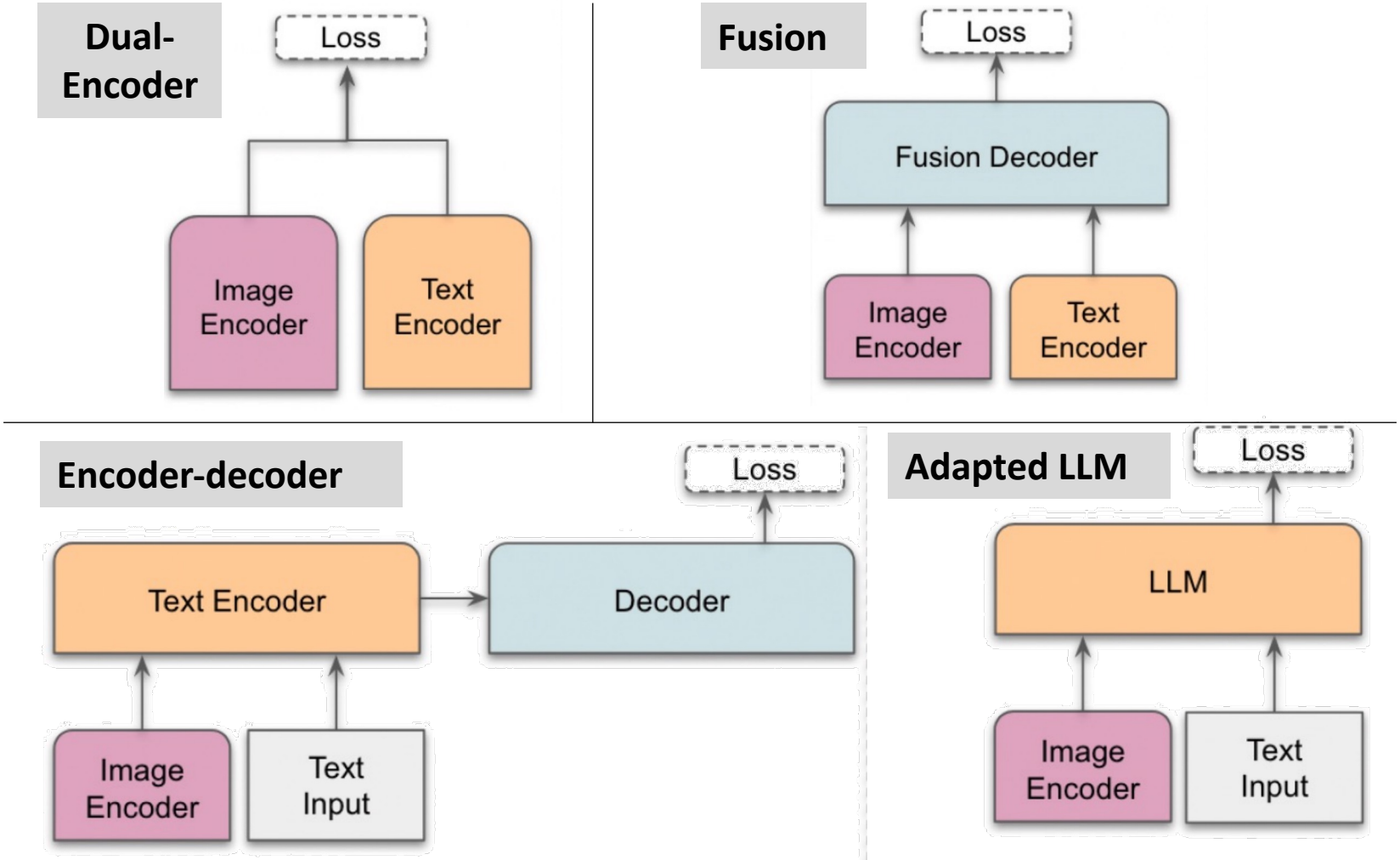
CLIP – Contrastive Language-Image Pretraining

- Inference

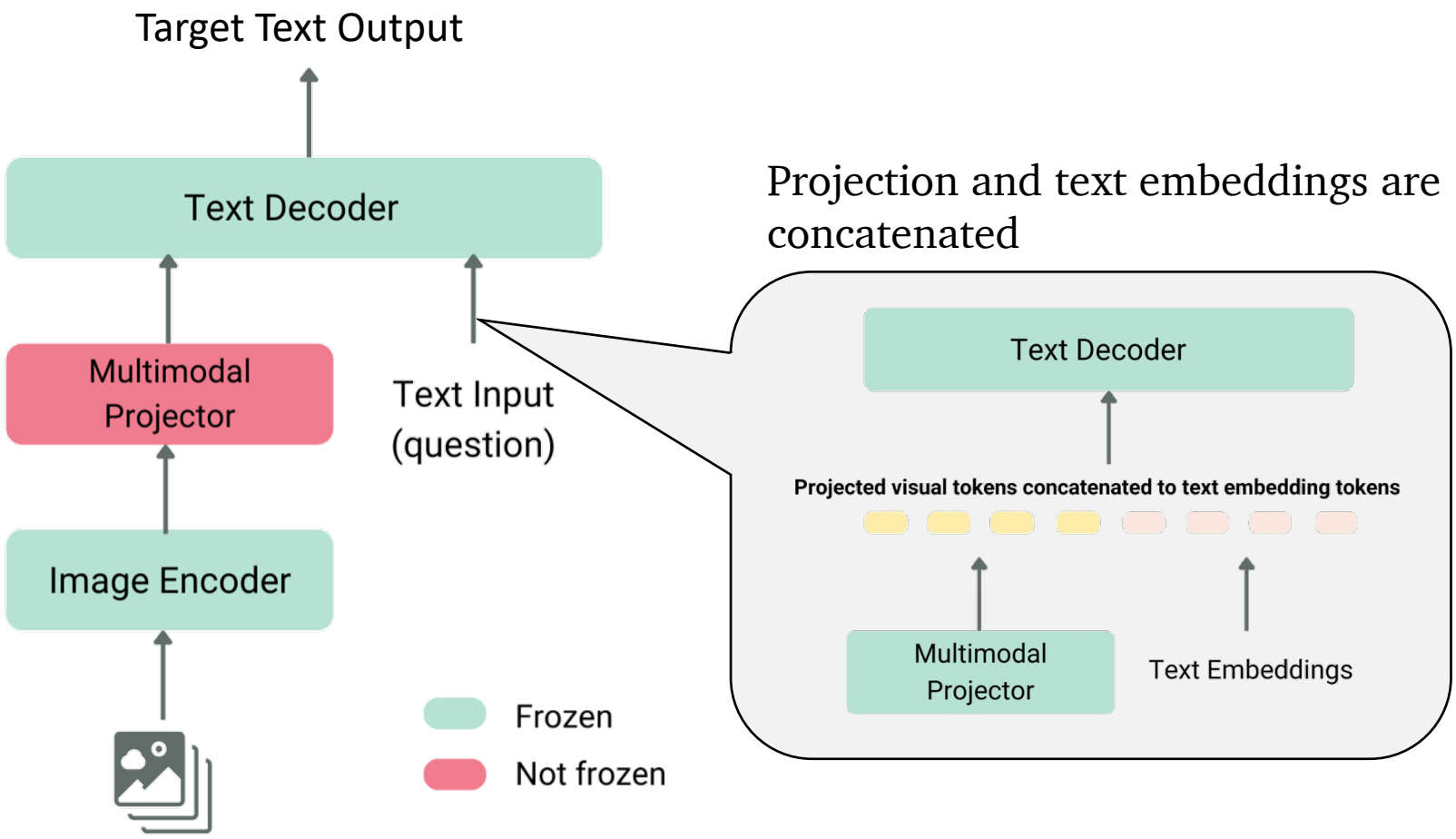


Learning Transferable Visual Models From Natural Language Supervision

Architectures of Vision-Language Models

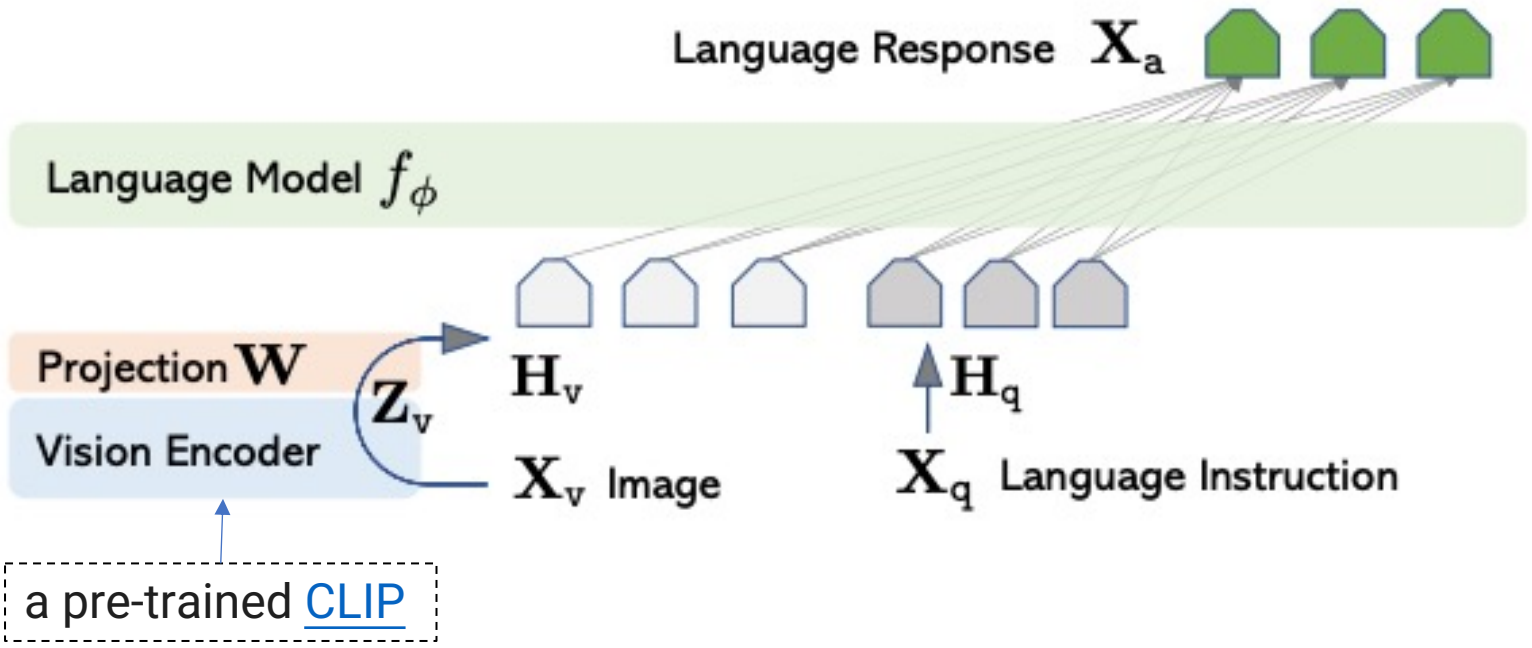


VLM with Adapted LLM



LLaVA

- Extends large language models like LLaMa with visual inputs

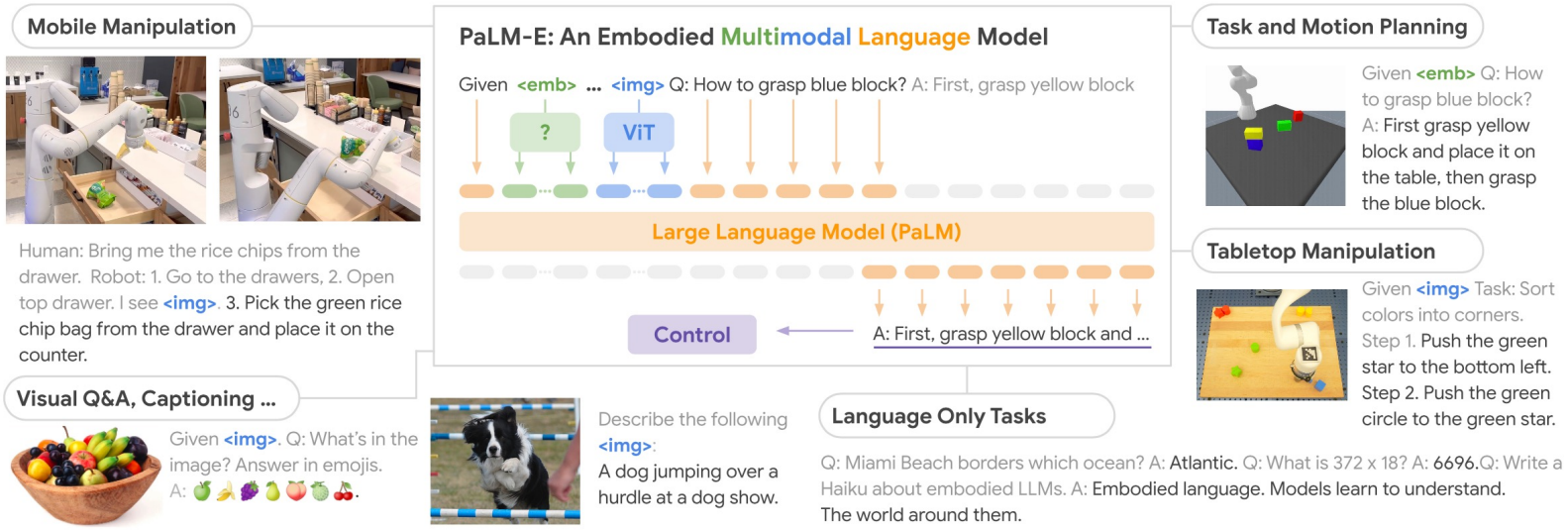


Popular VLMs

Model	Permissive License	Model Size	Image Resolution	Additional Capabilities
LLaVA 1.6 (Hermes 34B)	✅	34B	672x672	
deepseek-vl-7b-base	✅	7B	384x384	
DeepSeek-VL-Chat	✅	7B	384x384	Chat
moondream2	✅	~2B	378x378	
CogVLM-base	✅	17B	490x490	
CogVLM-Chat	✅	17B	490x490	Grounding, chat
Fuyu-8B	❌	8B	300x300	Text detection within image
KOSMOS-2	✅	~2B	224x224	Grounding, zero-shot object detection
Qwen-VL	✅	4B	448x448	Zero-shot object detection
Qwen-VL-Chat	✅	4B	448x448	Chat
Yi-VL-34B	✅	34B	448x448	Bilingual (English, Chinese)

More Modalities

- Embodied Multimodal Language Model



PaLM-E结合了PaLM-540B和ViT-22B, 可将文本、图像和机器人状态作为输入，编码进和单词嵌入的同一空间，然后输入到语言模型进行下一个单词的预测。

PaLM-E: An Embodied Multimodal Language Model https://palm-e.github.io/

TIME for Coding



- **Tutorial: Visual QA with LLaVa**

  https://colab.research.google.com/github/NielsRogge/Transformers-Tutorials/blob/master/LLaVa/Inference_with_LLaVa_for_multimodal_generation.ipynb#scrollTo=PuWVAAOinC8q

  https://huggingface.co/blog/vlms

注意:

```
inputs = processor(prompts, images=[image1,image2],
padding=True, return_tensors="pt").to("cuda")
```

需改成

```
inputs = processor(prompts[0], image1,
padding=True, return_tensors="pt").to("cuda")
```

What's Next?

- **Unsupervised Learning**



Clustering

Learn hidden structures

- K-means
- Gaussion Mixture



Generative

density estimation

- GAN
- VAE
- Diffusion



Dimensional Reduction

Learning data compression

- Principal component analysis
- Auto-encoder

